Příprava monokrystalů pro materiálový výzkum a aplikace

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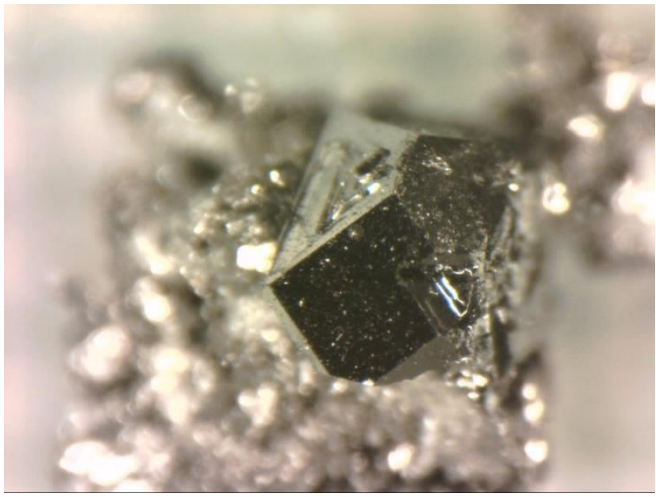
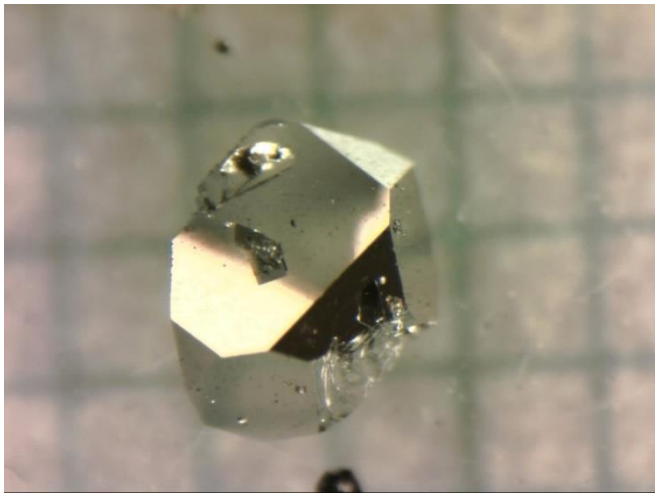
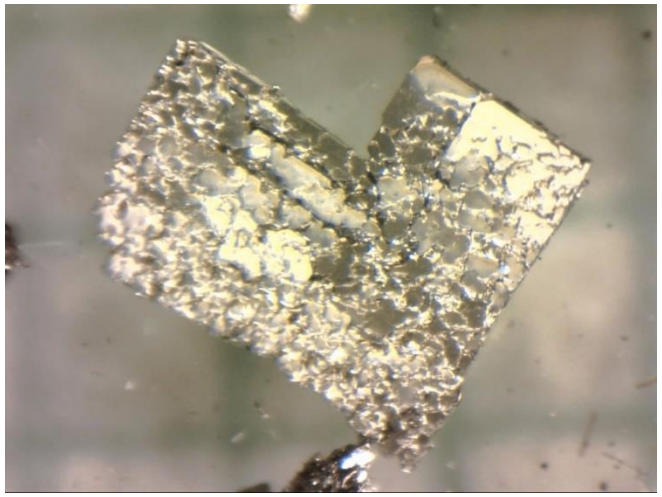
Katedra fyziky kondenzovaných látek

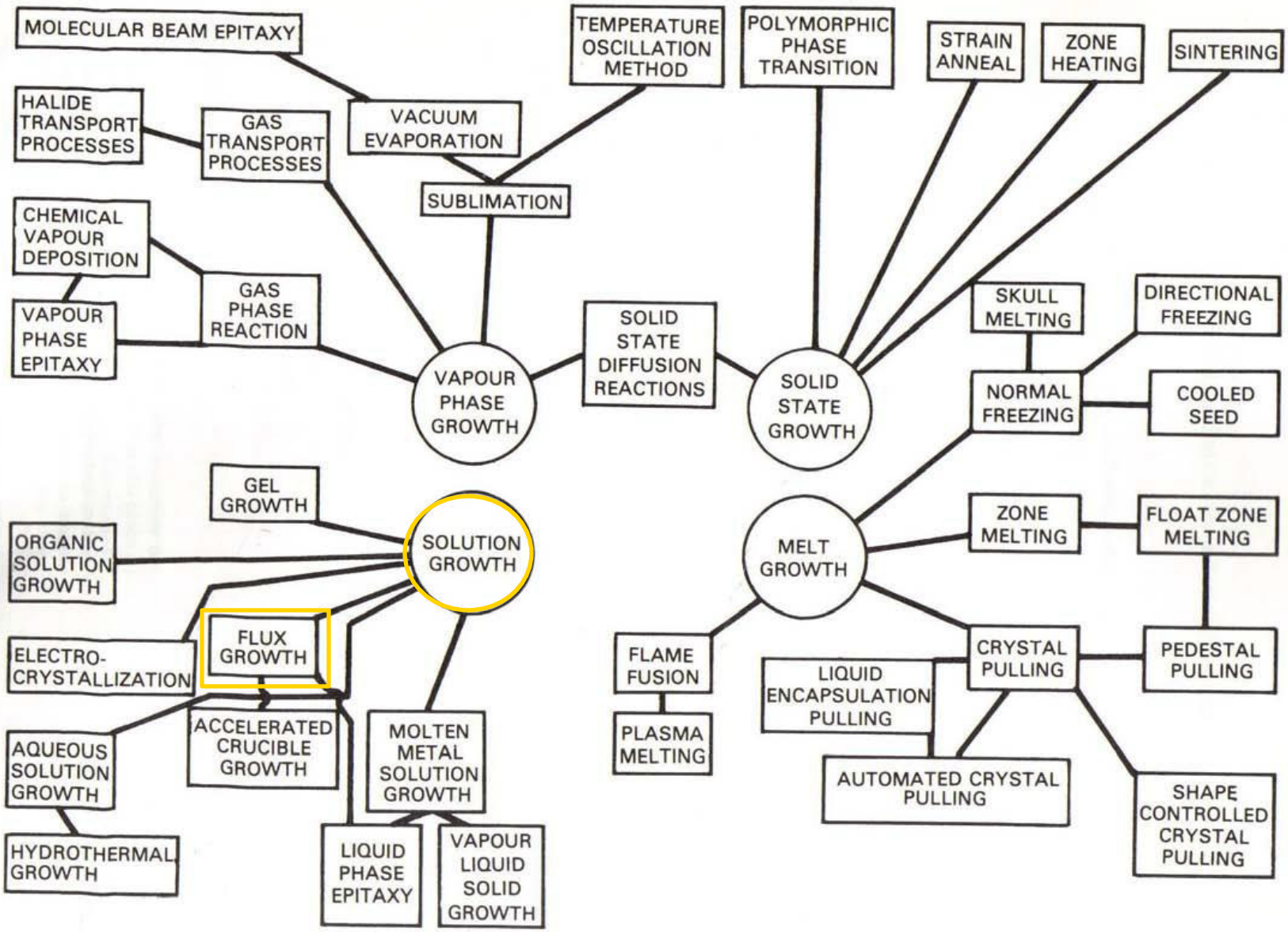
Flux Method







What are the crystal growth techniques?



B. R. Pamplin, Crystal Growth, Pergamonpress (1980).

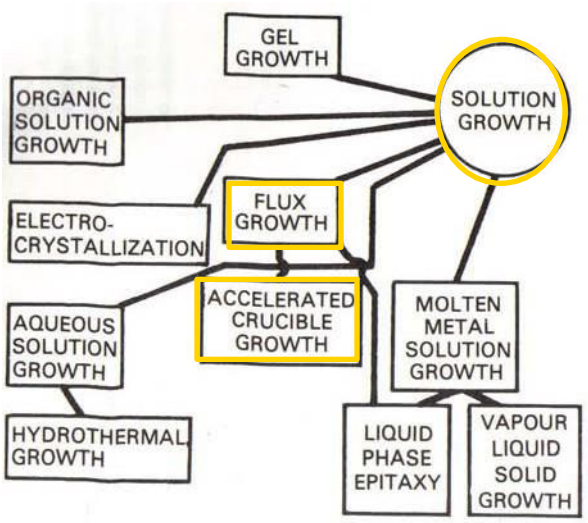
✐ Why solution growth?

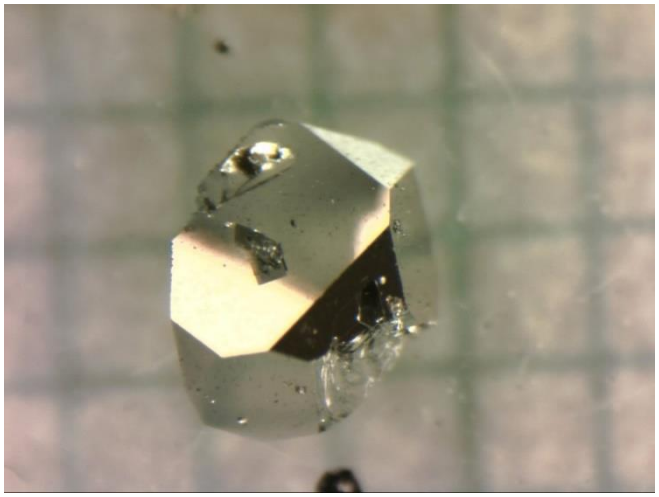
Advantages:

- Grow congruently and incongruently melting materials
- Need relatively simple equipment-cheap
- Has relatively short growth-time scales
- Need small amounts of materials
- Single crystals of very high quality

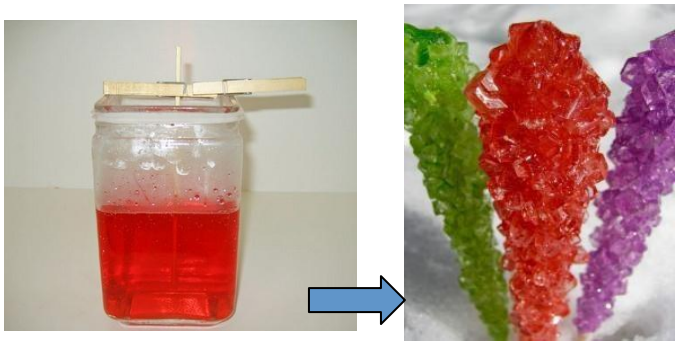
Disadvantages:

- Not too large crystals (mm to cm)
- **No recipe how to growth new materials**





Classical solution vs. flux growth?



- Evaporation of the solvent

- Solubility as a function of temperature

▪**Flux Growth** is solution growth at high temperature, oversaturated solutions

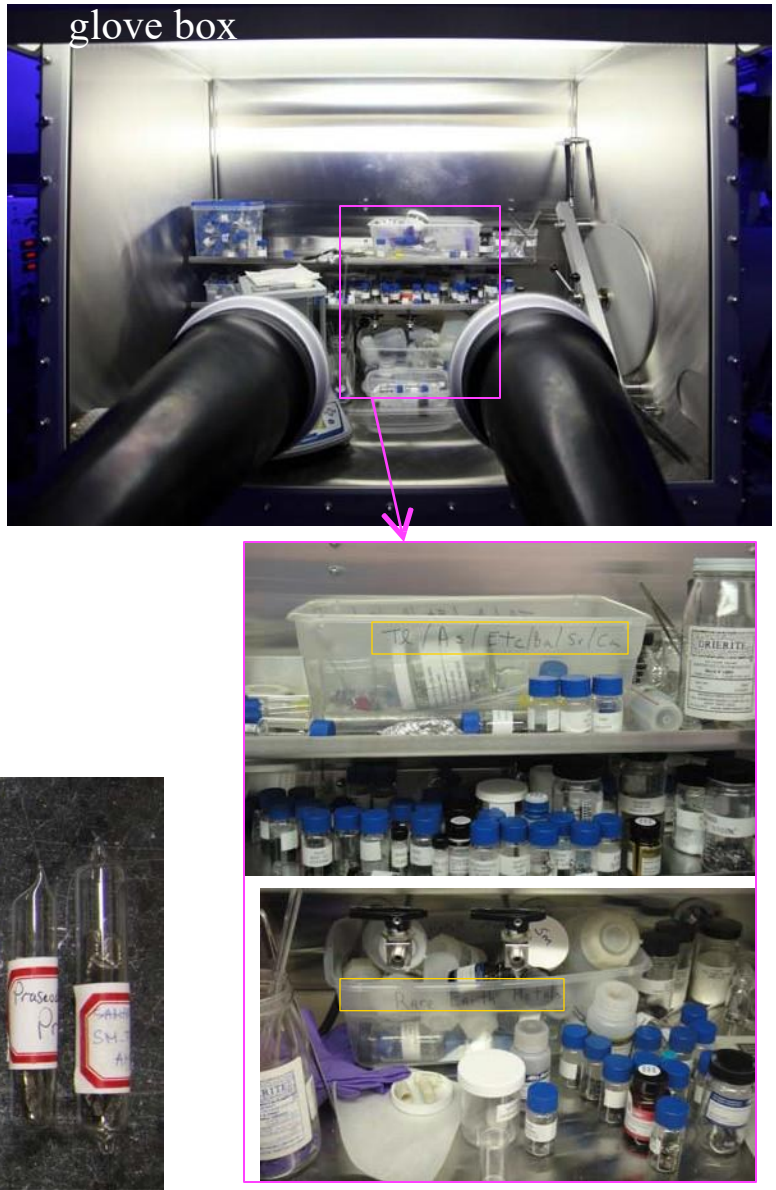
▪Flux (melt; solvent) can be metals (Ni, Fe, etc.), oxides (B_2O_3, Bi_2O_3), hydroxides (KOH, NaOH), salts (BaO, PbO, PbF_2), eutectic binaries

Key characteristics for fluxes

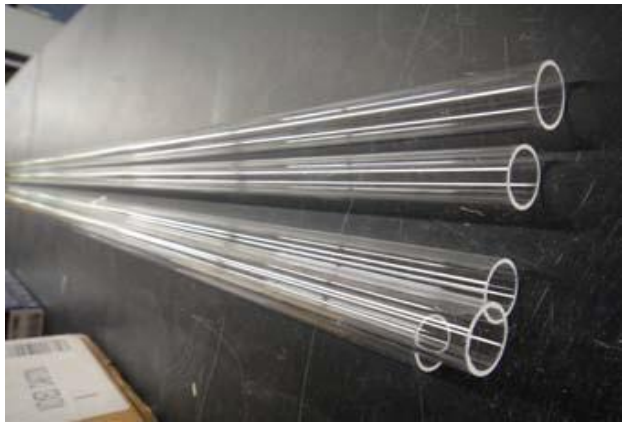
- Have low melting temperature
- Be easily separated from the products
- (Not form stable compounds with the reactants)-self flux!
- Have a large difference between boiling & melting temp.

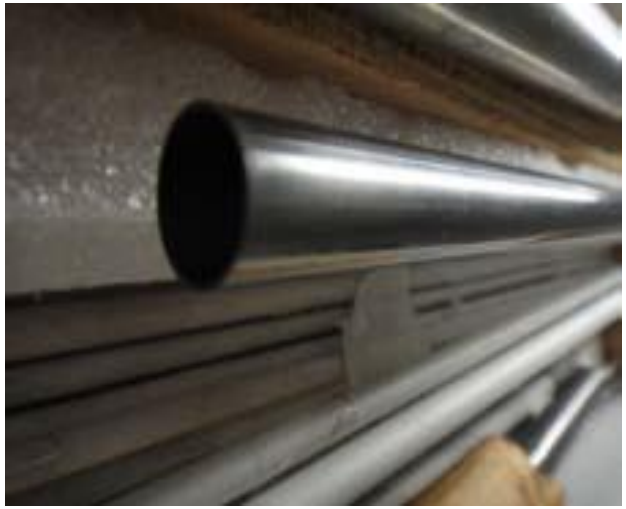
What flux growth needs?

- Elements
- Cutting tools



What flux growth needs?



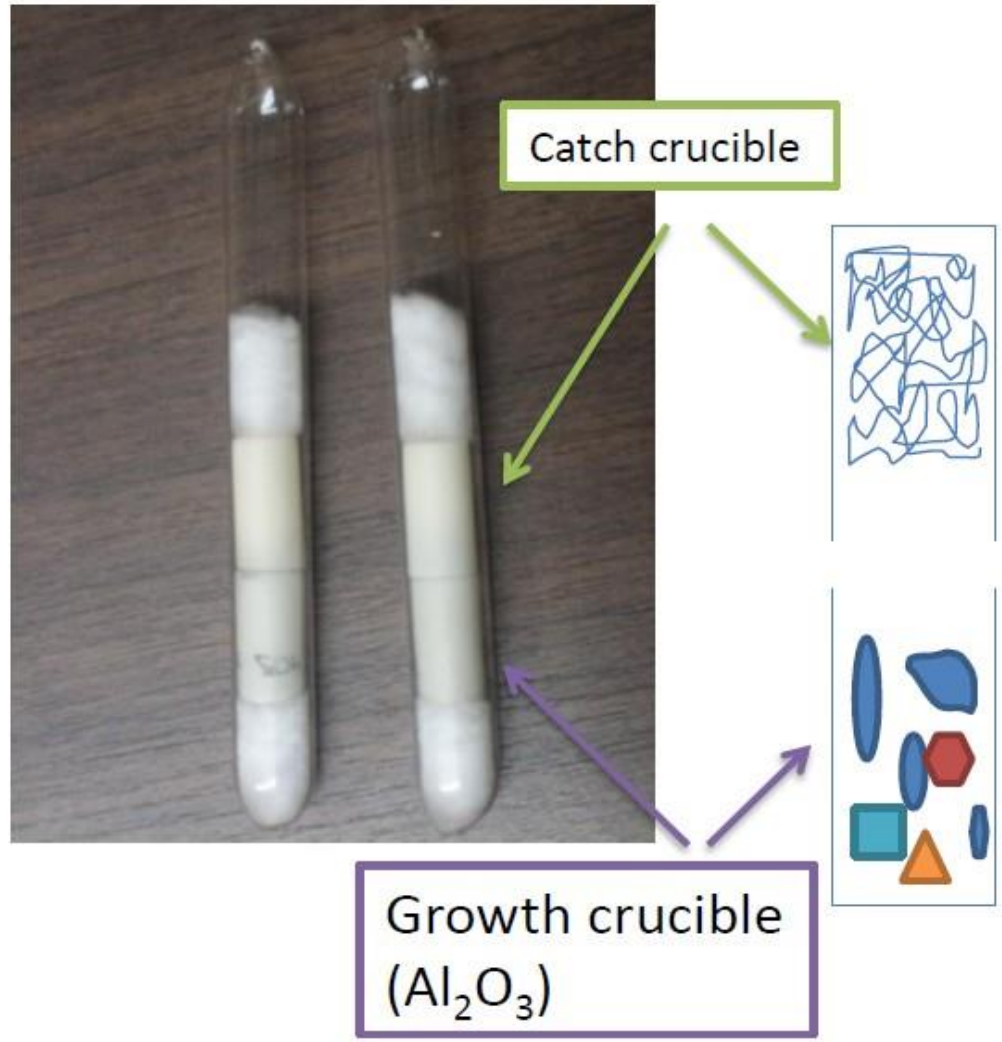


- Crucibles-reaction with the flux or components
- Tubes (reaction under ambient conditions not possible)

	T_{max} (°C)	$T_{melting}$ (°C)
borosilicate glass (Pyrex)	515	820
gold	1013	1064
silica (quartz)	1200	1853
platinum	1720	1770
alumina (Al_2O_3)	1900	2072
zirconia (ZrO_2)	2000	2700
magnesia (MgO)	2400	2852
tantalum	1400	3017

What flux growth needs?

- Glass sealing station



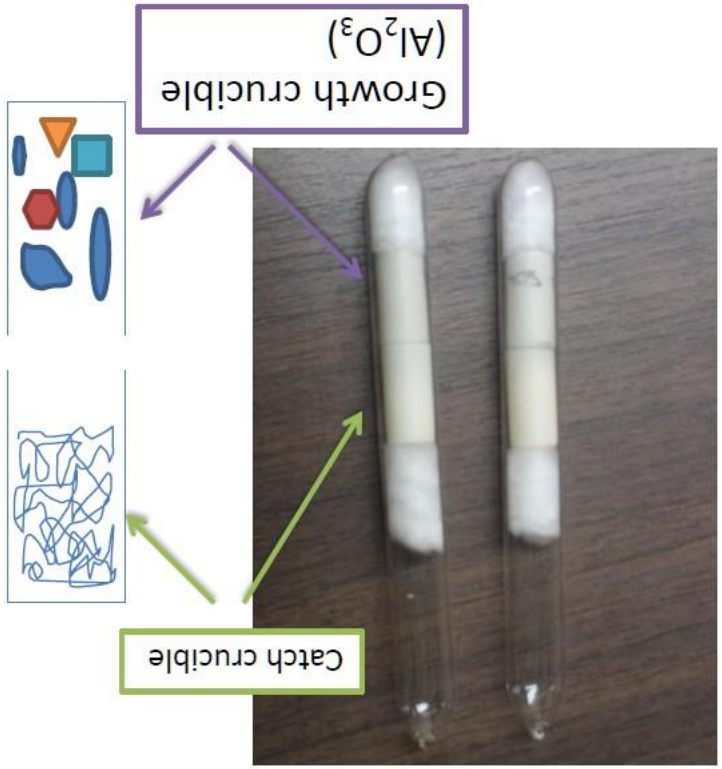
What flux growth needs?

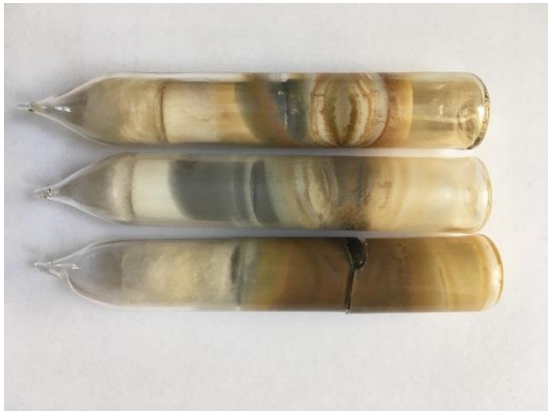
- Furnaces

SiC heating elements (T_{max}= 1500 °C)
$MoSi_2$ heating elements (T_{max} = 1700 °C)

What flux growth needs?

- Centrifuges



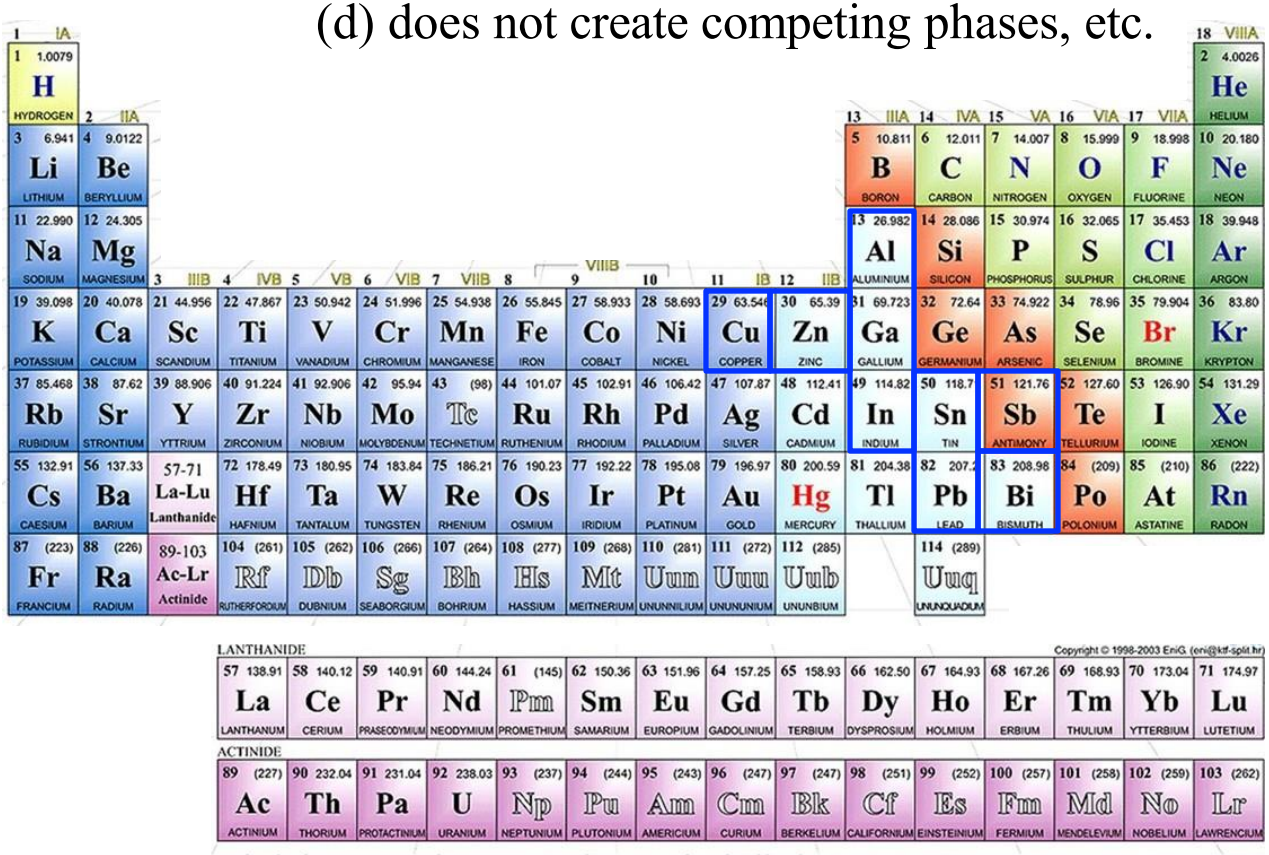


- Chemical etchers (e.g. for Al, use NaOH; for Ga or In, use HCl)
- Mechanical removal tools

Where to start?

❖ (5) Find a good metallic flux for crystal growth

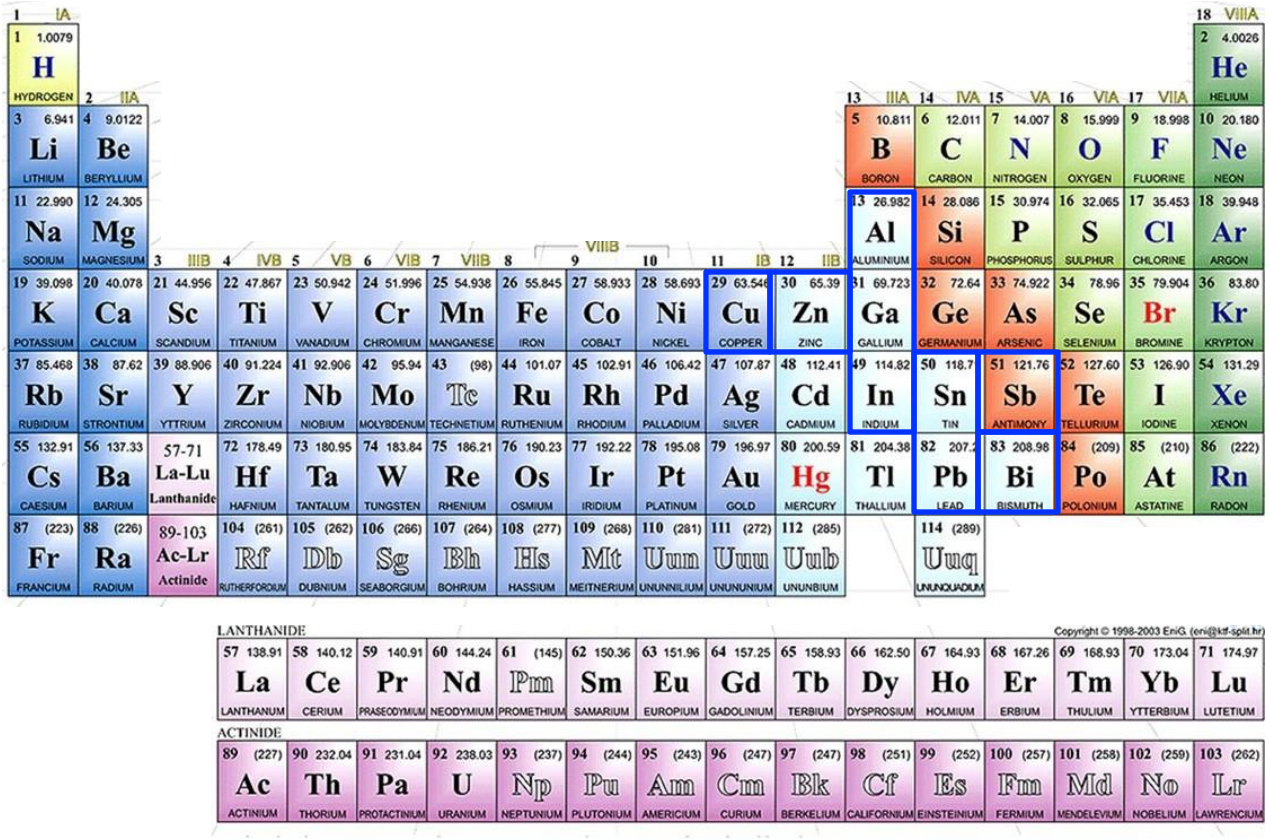
A good flux (a) has a low melting temperature,
(b) has a good solubility for the elements,
(c) does not enter crystal as inclusions,
(d) does not create competing phases, etc.



Review of some metallic fluxes

Al
- T_{melt} = 660 °C
- attacks silica
- spin off Al in silica tubes or use NaOH

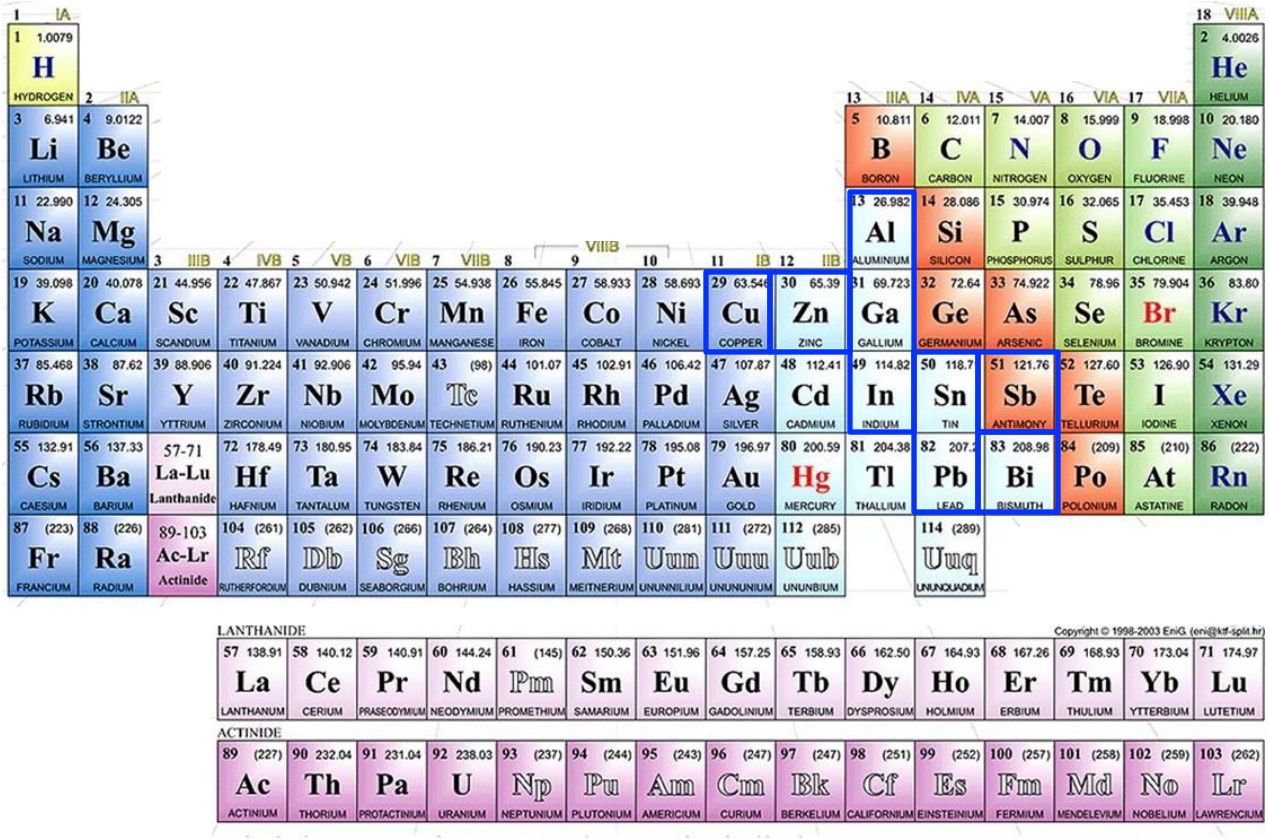
e.g. RB_4, $YbAlB_4$, RB_6, RBe_{13}, RAl_3, TiB_2, $CeSi_{2-x}$



Review of some metallic fluxes

Ga
- $T_{melt} = 30\ °C$
- tends to wet surfaces of grown crystal
- forms compounds with rare-earths

e.g. RSb, $R_2Pt_4Ga_8$

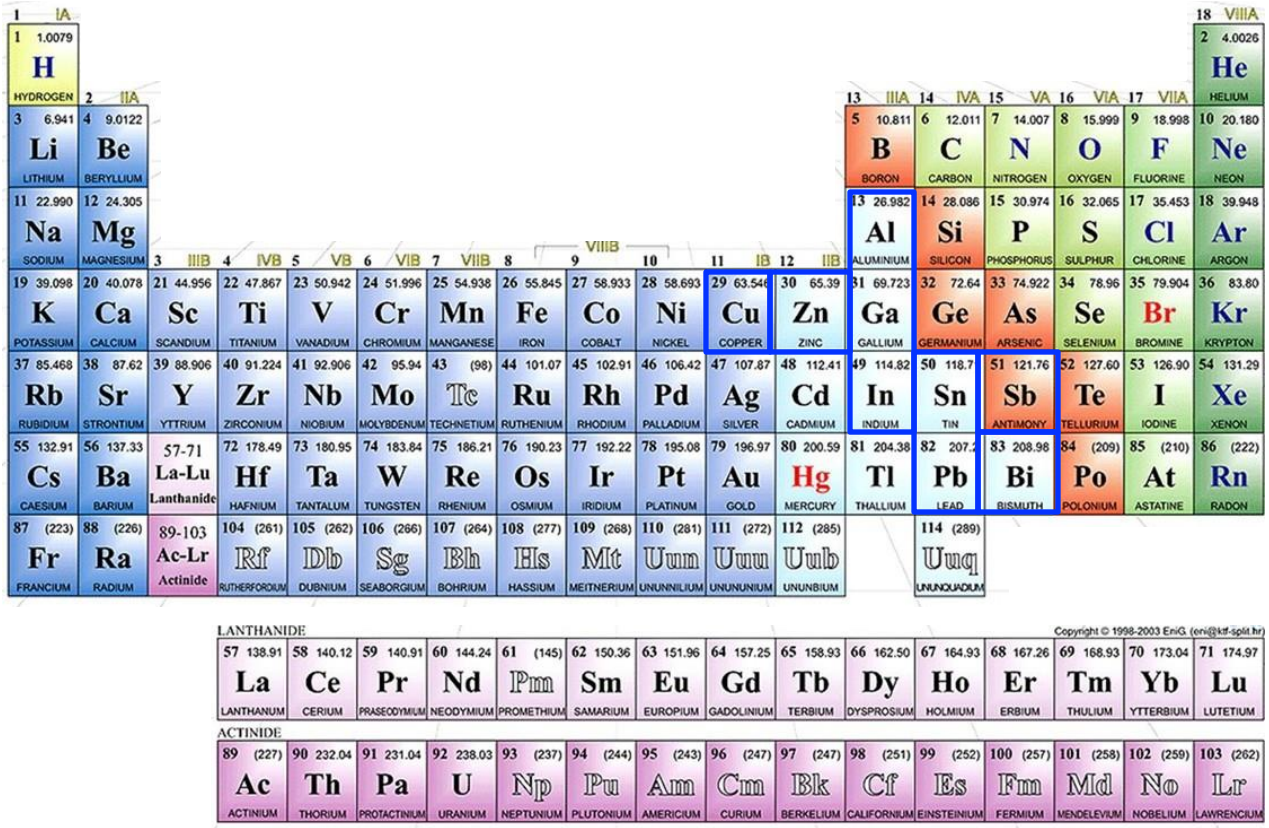


Review of some metallic fluxes

In • $T_{melt} = 157\ °C$

- Proven to be a good flux for $ThCr_2Si_2$-types
- $T_C = 3.4\ K$

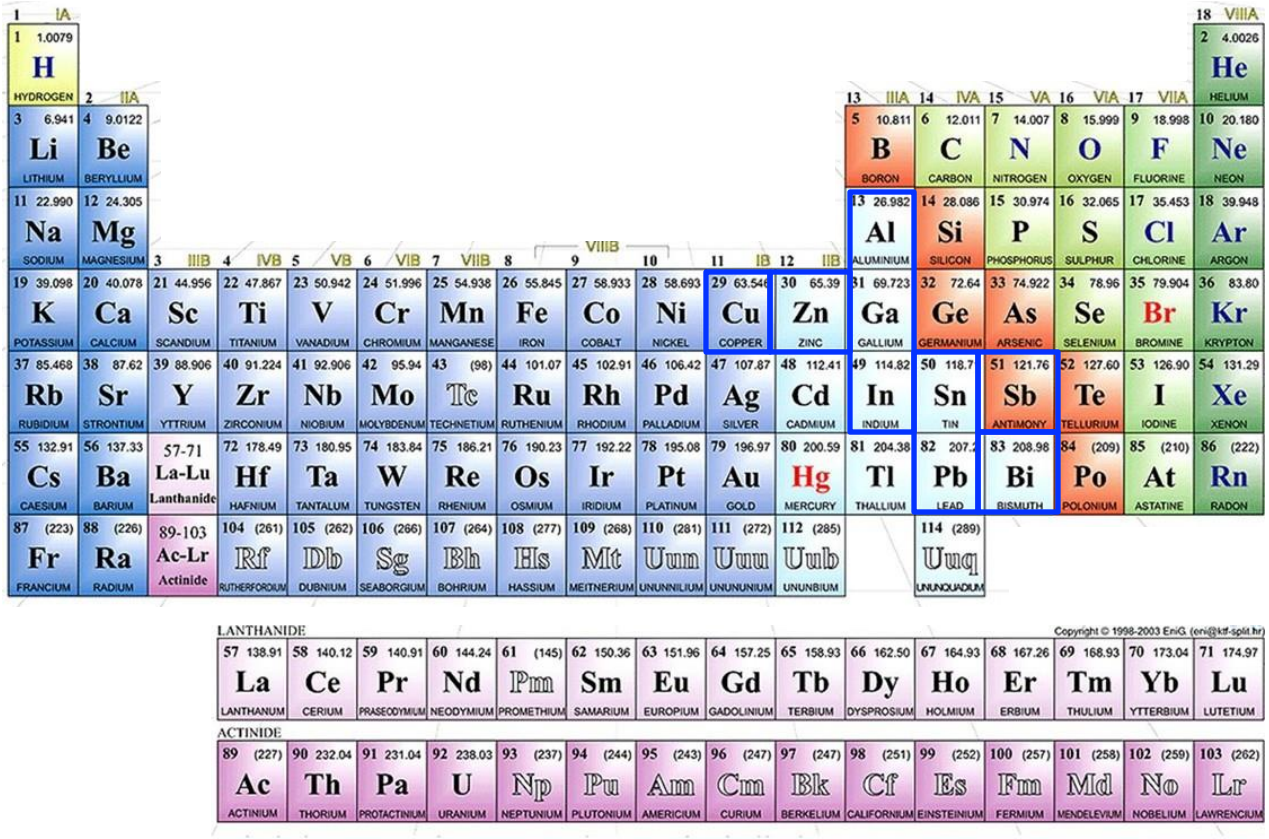
e.g. $CeCu_2Ge_2$, $CeNi_2Ge_2$, $TyCu_2Si_2$, Ce_3PdIn_{11}



Review of some metallic fluxes

Sn; Pb
- T_{melt} = 232 °C; 327 °C
- form RPb_3, RSn_3 phases
- T_C = 3.7 K; 7.2 K

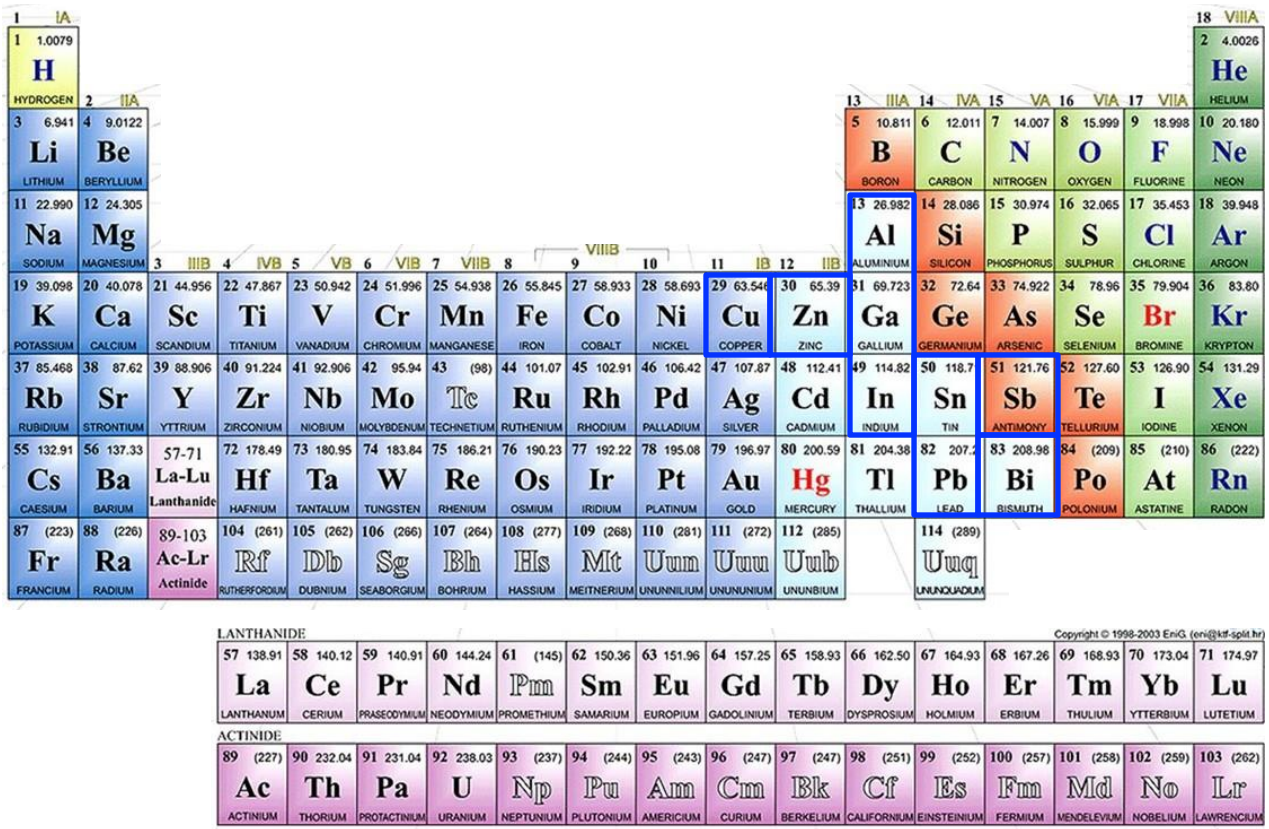
e.g. $YbCu_2Si_2$, TiNiSn, MnSnNi, *R*Sb; *R*BiPt, RPbPt



Review of some metallic fluxes

Sb
- $T_{melt} = 630$ °C
- stability of *R*-Sb phases

e.g. RSb_2, $U_3Sb_4Pt_3$, $PtSb_2$

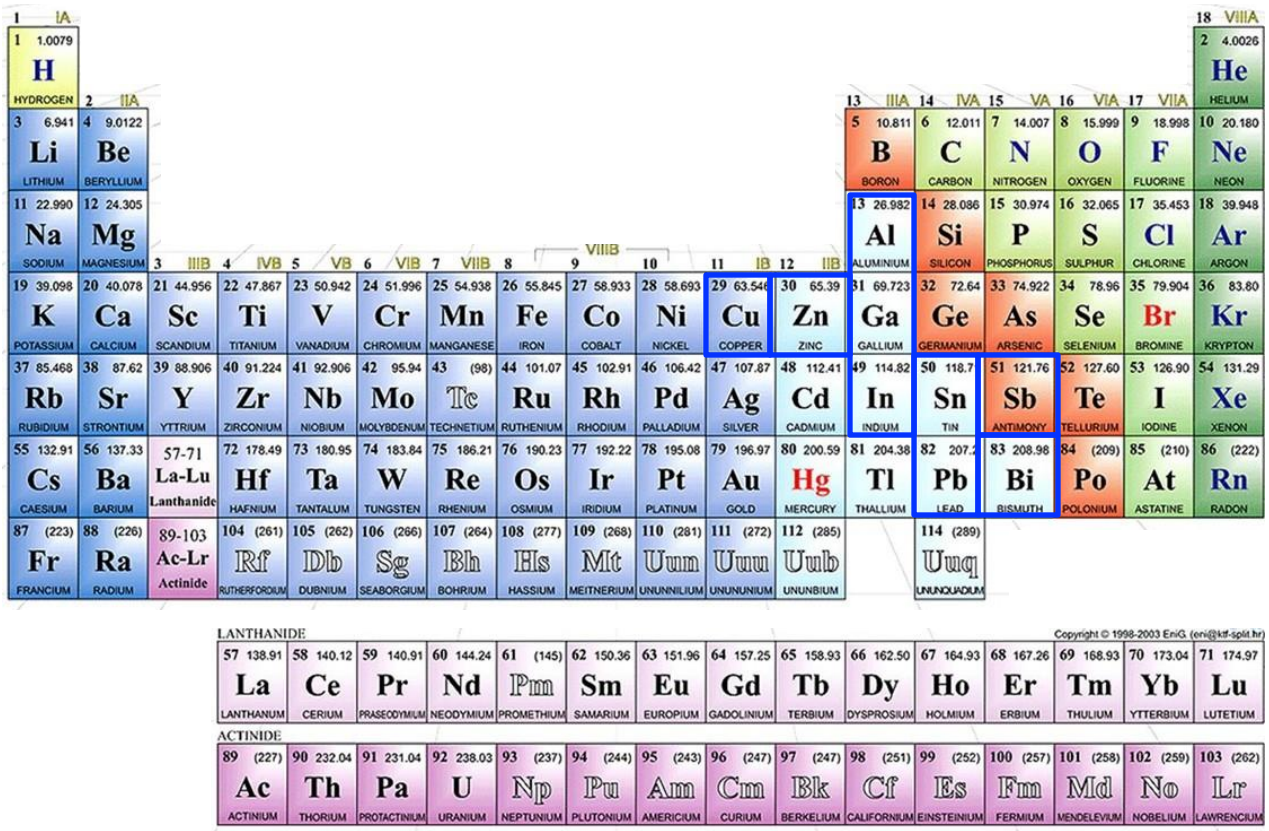


Review of some metallic fluxes

Zn $T_{melt} = 420$ °C e.g. InSb, GaSb, InAs, Si, Ge

Bi $T_{melt} = 272$ °C e.g. UPt_3, PtMnSb, NiMnSb, UAl_3, GaP, $ZnSiP_2$, $CdSiP_2$

Cu $T_{melt} = 1085$ °C e.g. RPh_4B_4, RCu_2Si_2, V_3Si, RIr_2, UIr_3



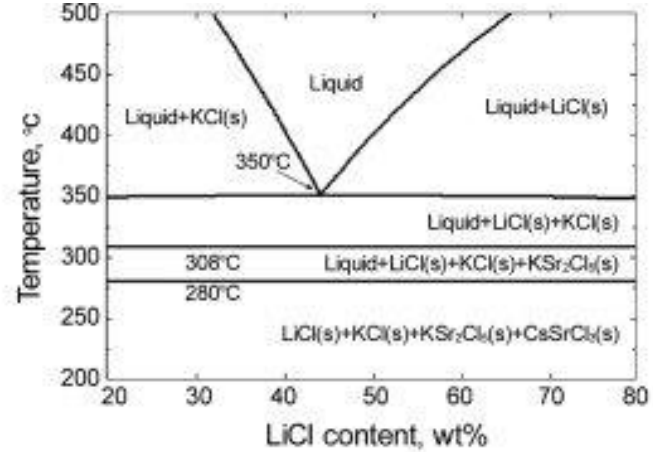
Other possible fluxes

Key characteristics for fluxes

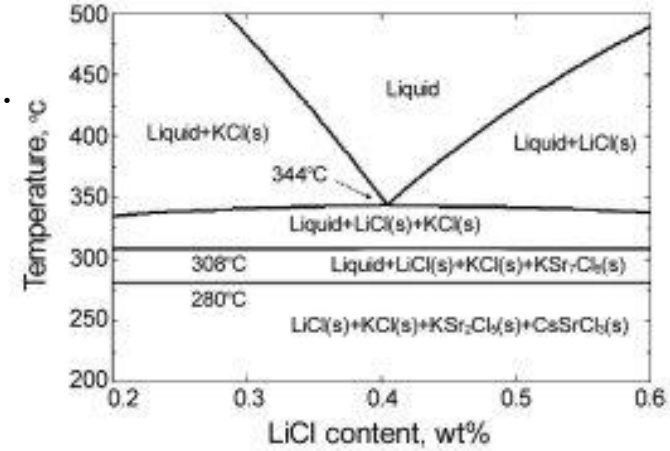
- Have low melting temperature
- Be easily separated from the products
- Not form stable compounds with the reactants
- Have a large difference between boiling & melting temp.

Generally inorganic salts can be used for fluxes

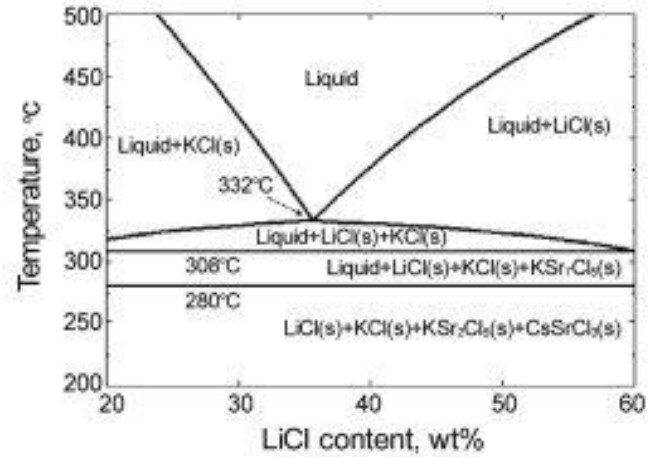
- **Pure compounds**
- **Eutectics**



(a) eutectic(LiCl-KCl) salt(98 wt%)-CsCl(1 wt%)-$SrCl_2$(1wt%)



(b) eutectic(LiCl-KCl) salt(90 wt%)-CsCl(5 wt%)-$SrCl_2$(5 wt%)



(c) eutectic(LiCl-KCl) salt(80 wt%)-CsCl(10 wt%)-$SrCl_2$(10 wt%)

✎ Where to start?

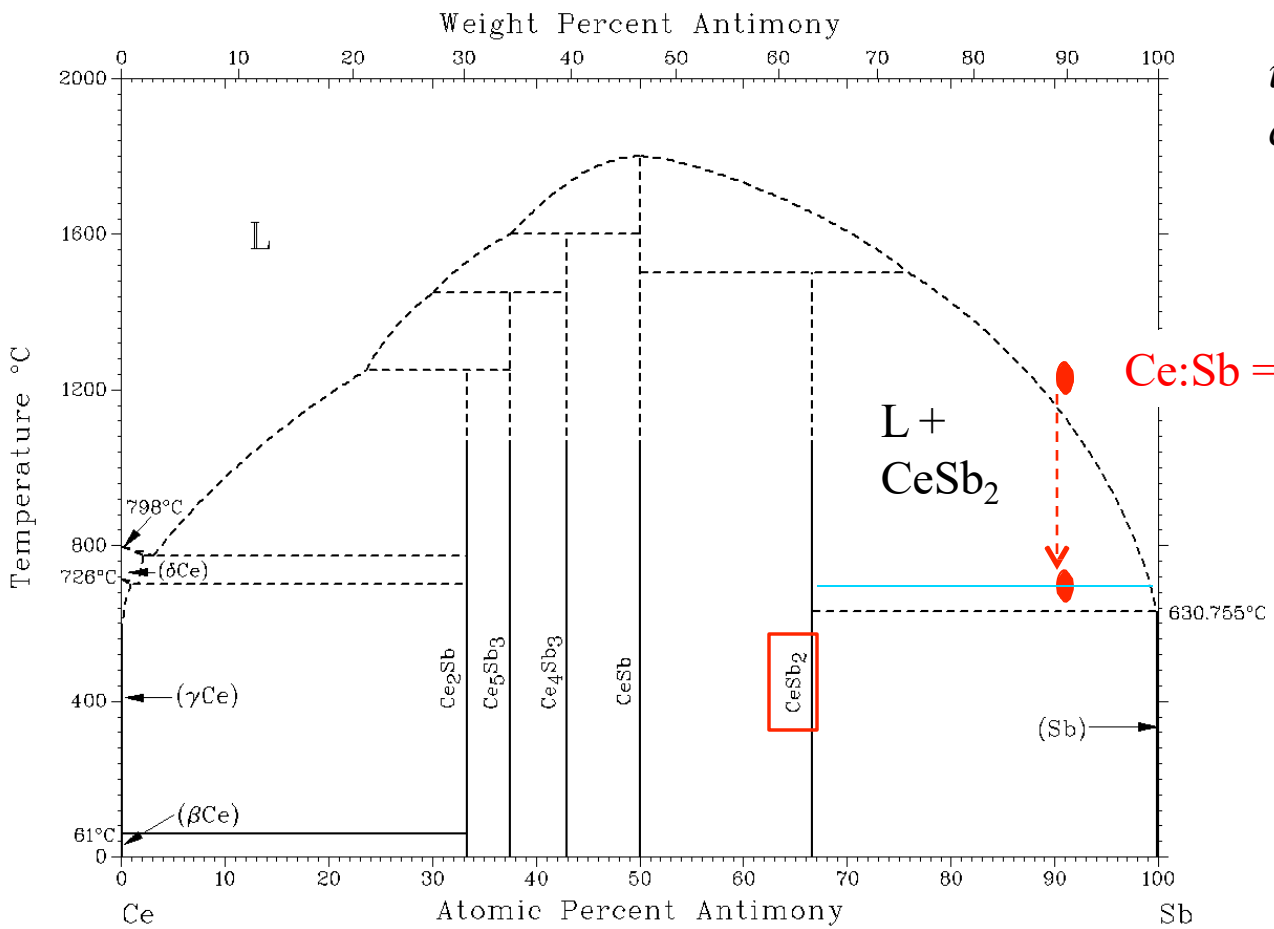
Find the existing phase diagrams

-Binary diagrams are a good start (may be your only option)

-Study of literature!!! – similar systems

Some examples

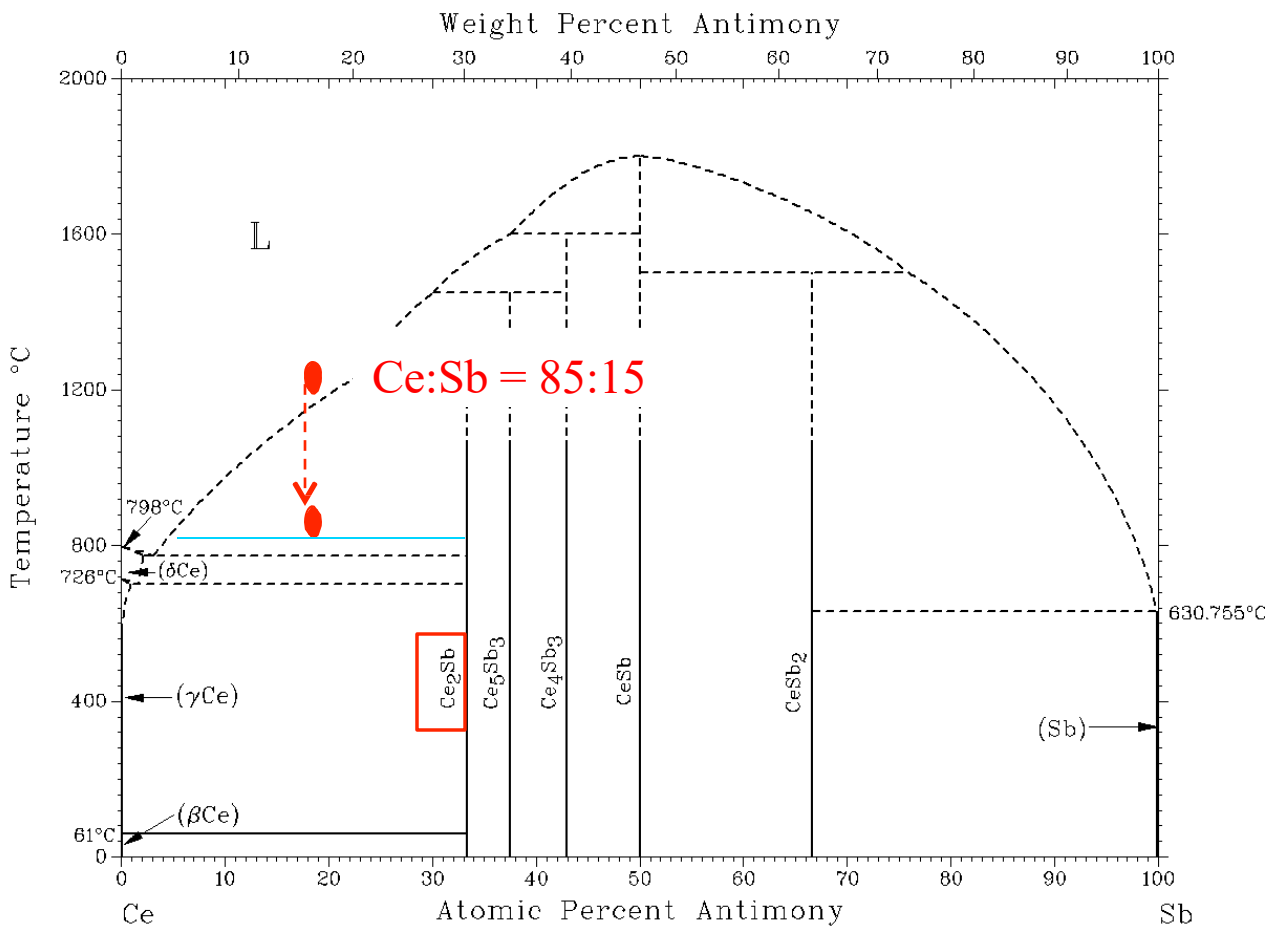
(2) How to make $CeSb_2$? Melts incongruently



use alumina crucible

Some examples

(3) How to make Ce_2Sb? Melts incongruently



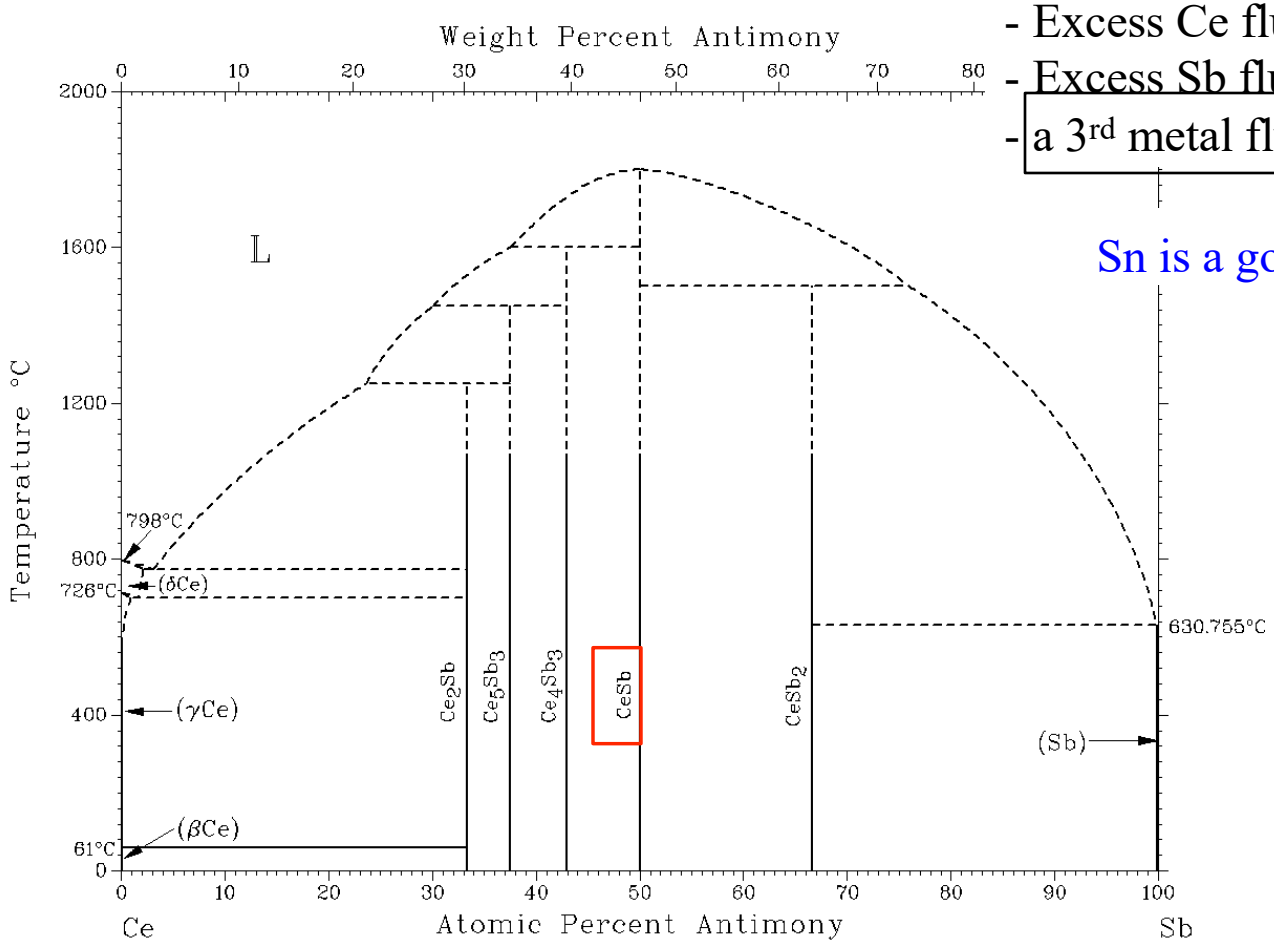
use Ta crucible

Some examples

(4) How to make CeSb? Melts congruently at 1820 °C

- Melt growth technique
- Excess Ce flux
- Excess Sb flux
- a 3rd metal flux?

Sn is a good choice

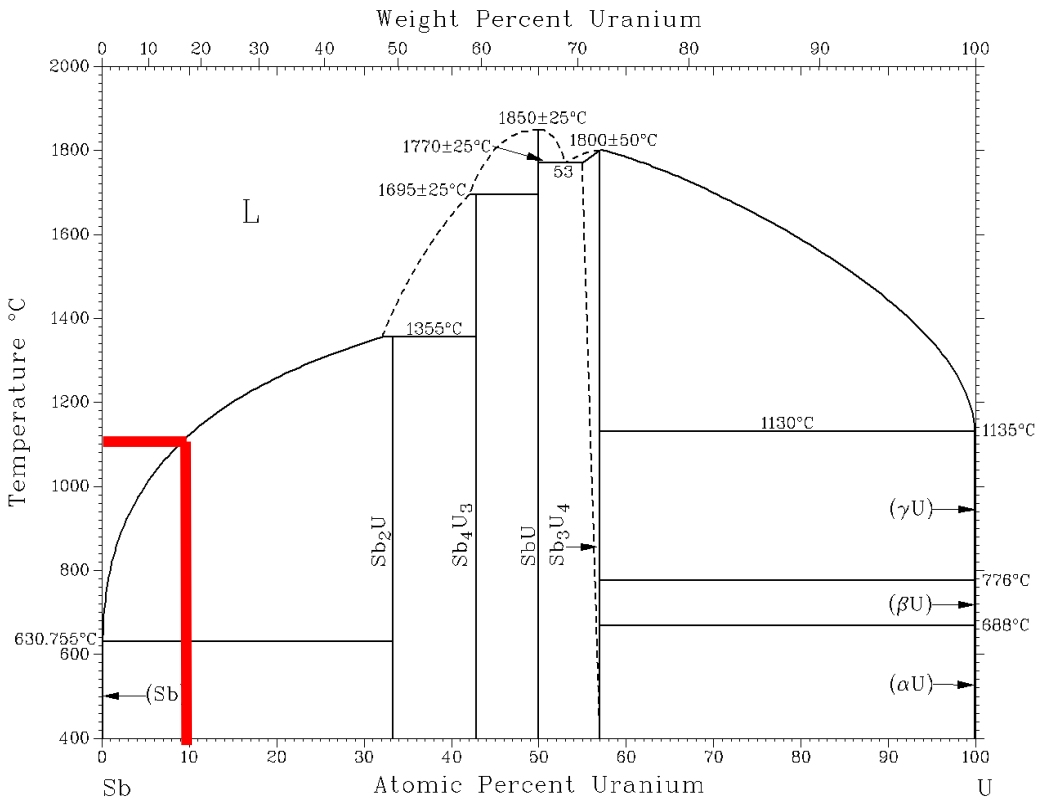


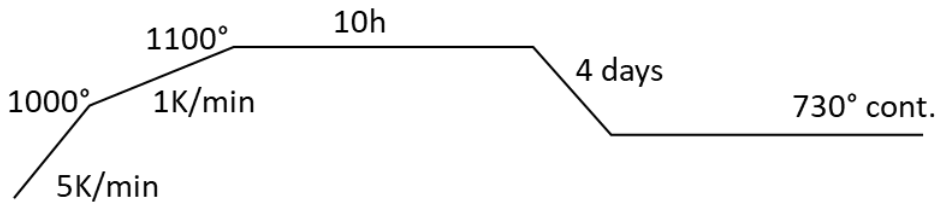
Growth of $U_3T_3Sb_4$

Self flux- Sb

☹ Too low content of Sb in formula

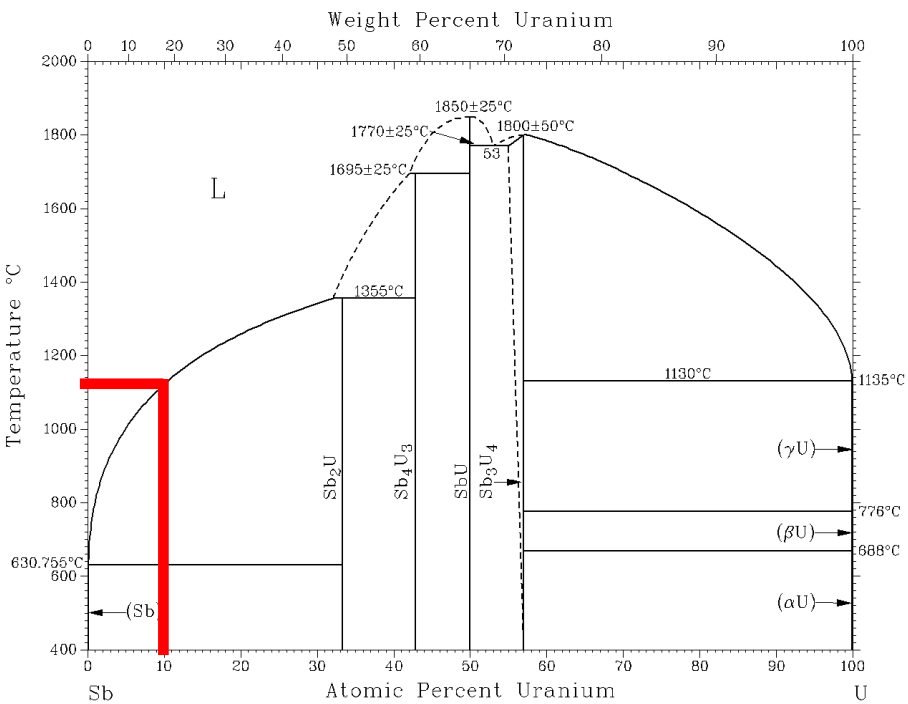
Thermal profile

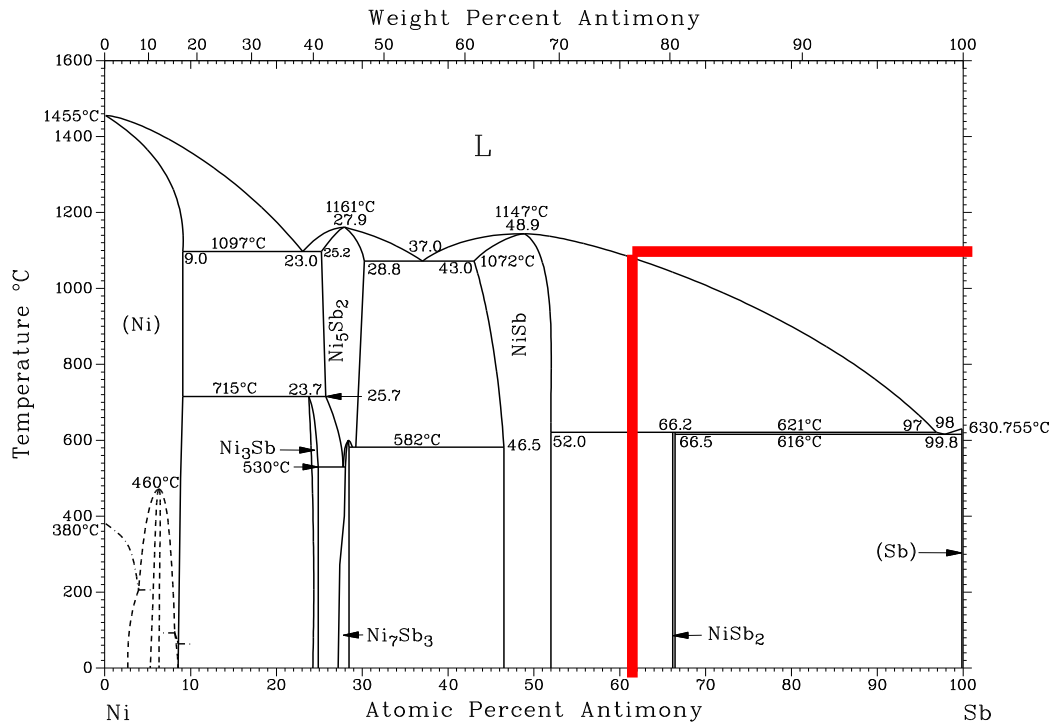




- Upper temperature 1100°C
  - Temperature limit of quartz
- Solubility of U only 10%

$U_3Ni_3Sb_4$

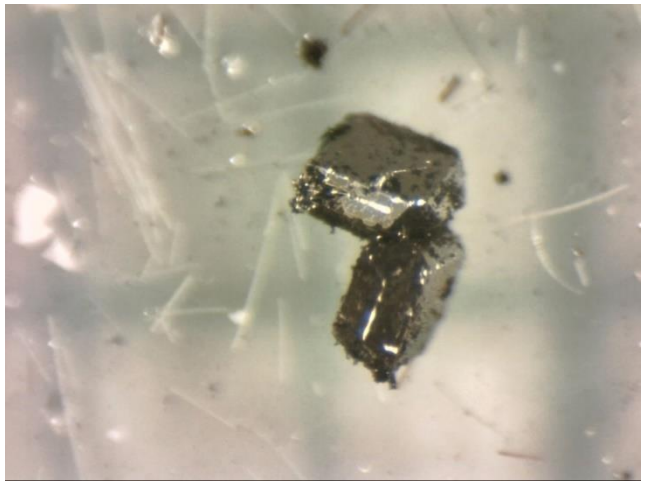


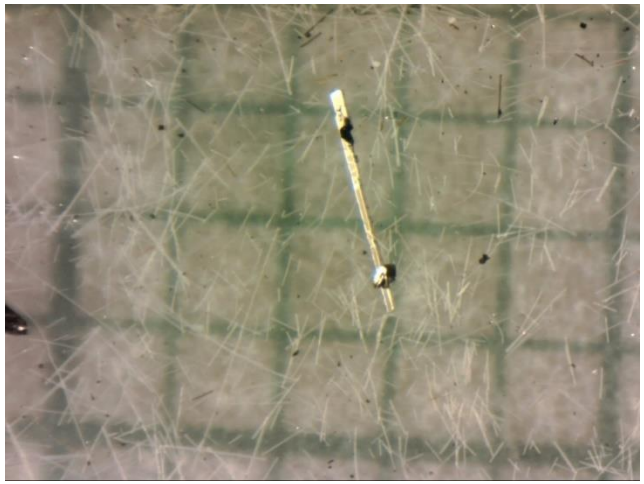


- Solubility of Ni in Sb-flux is significantly higher than U

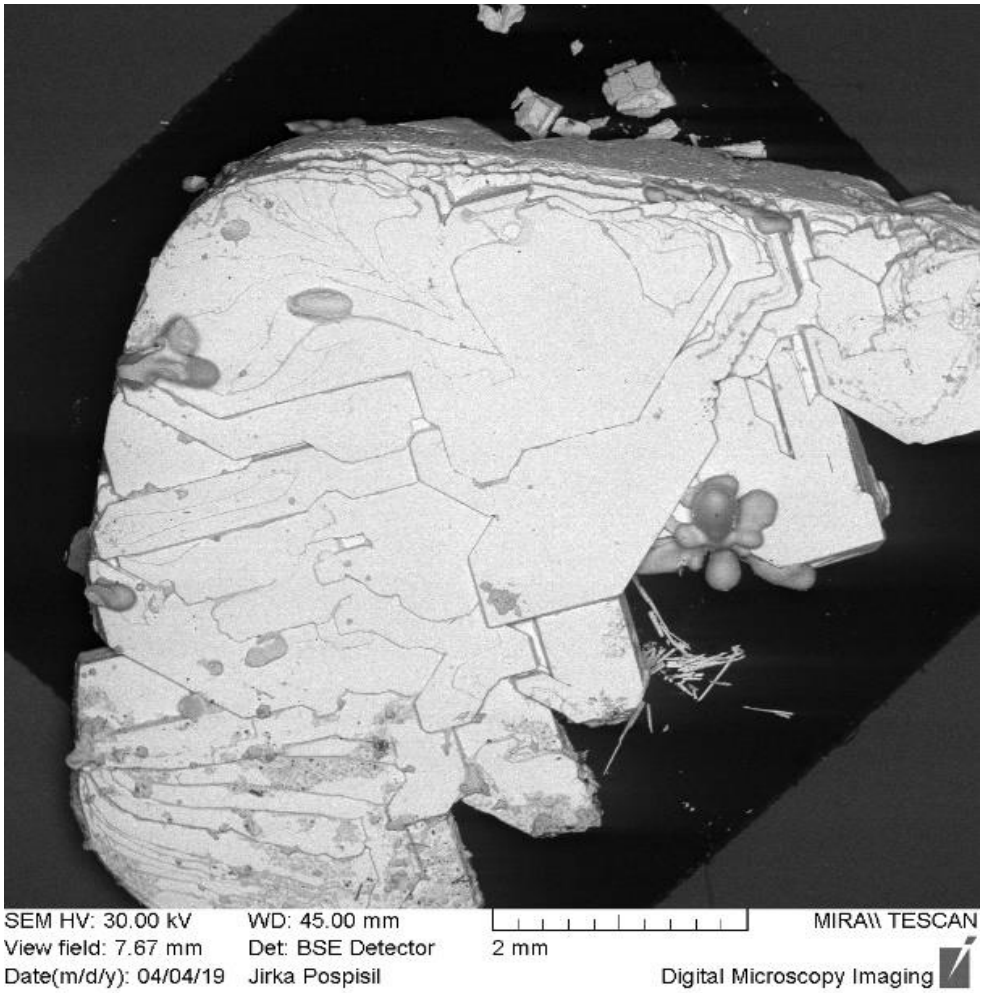
$U_3Ni_3Sb_4$

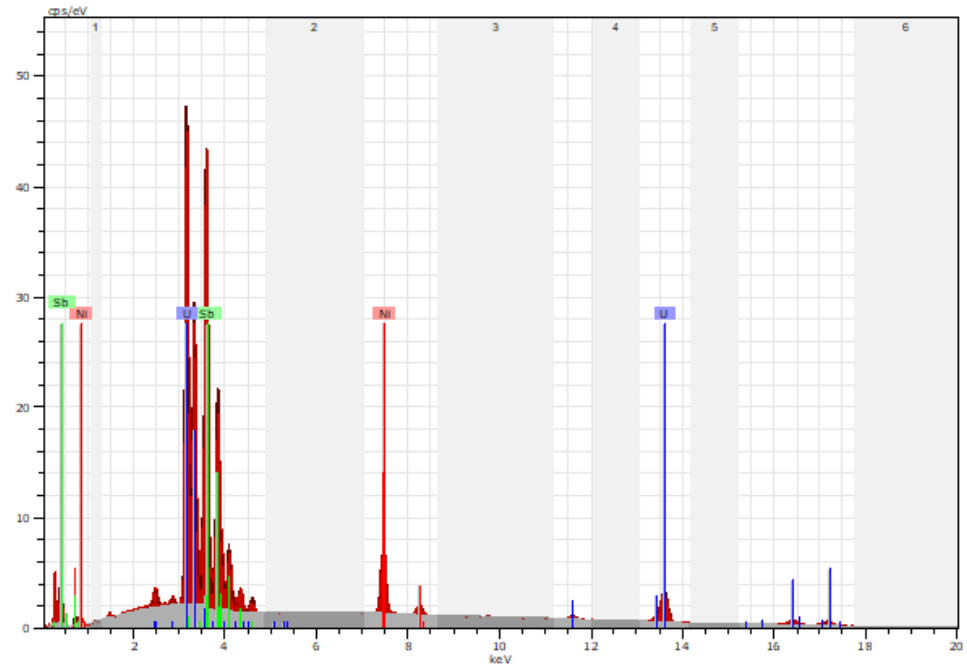
Element	Sb(6N)		Ni(4N5)		U(3N)
Ratio	10	:	2	:	1
Mass	4,36 g		0,43 g		0,84 g





$U_3Ni_3Sb_4$ - Chemical analysis

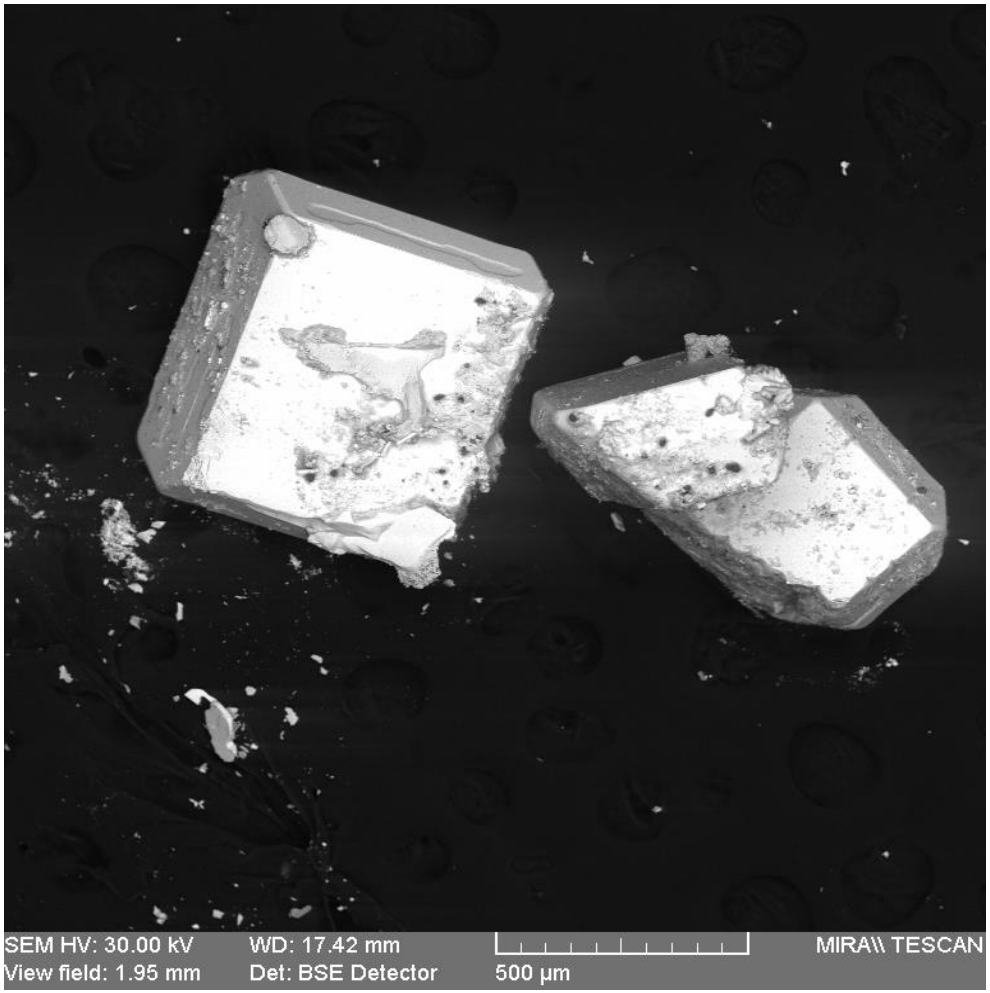


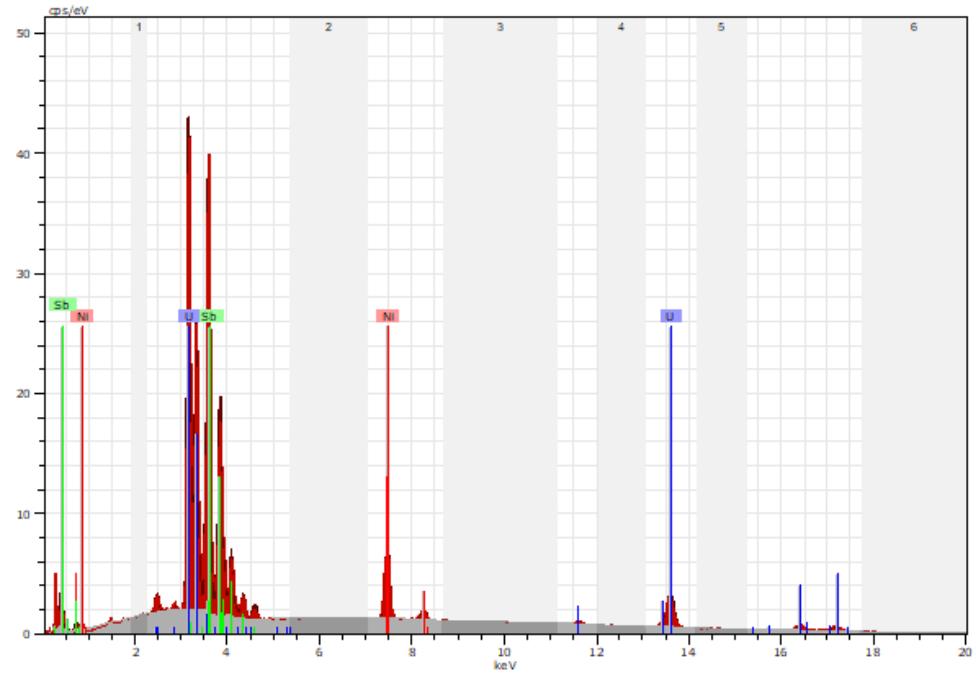


Element	Expected ratio	Final ratio
U	1	1
Ni	1	0,36
Sb	1,33	1,69

~U_3NiSb_5, unreported new phase

$U_3Ni_3Sb_4$ - Chemical analysis



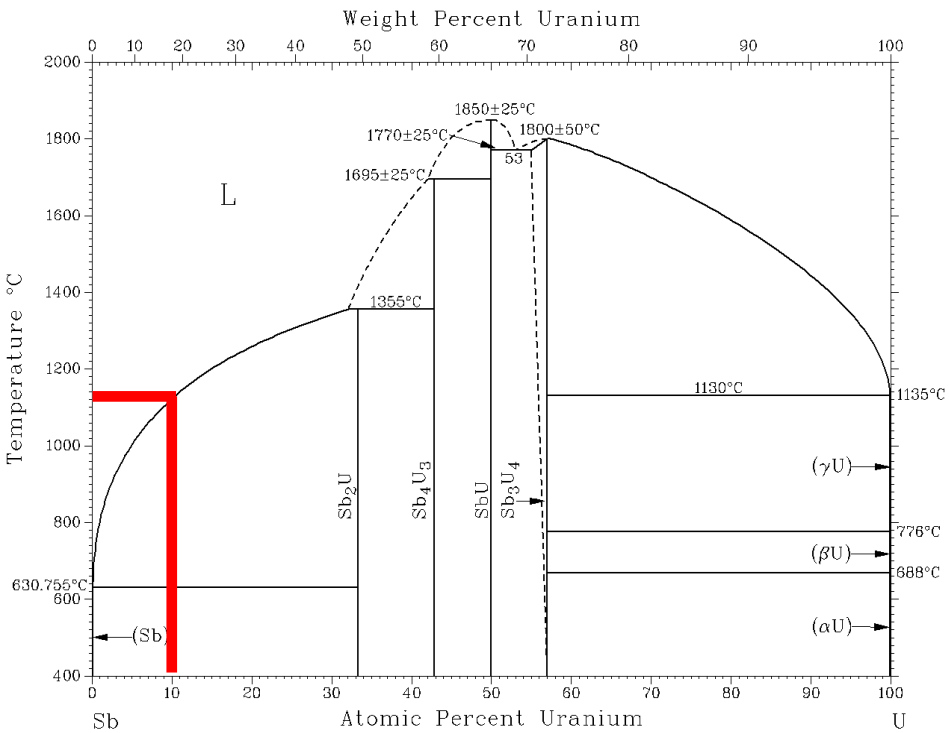


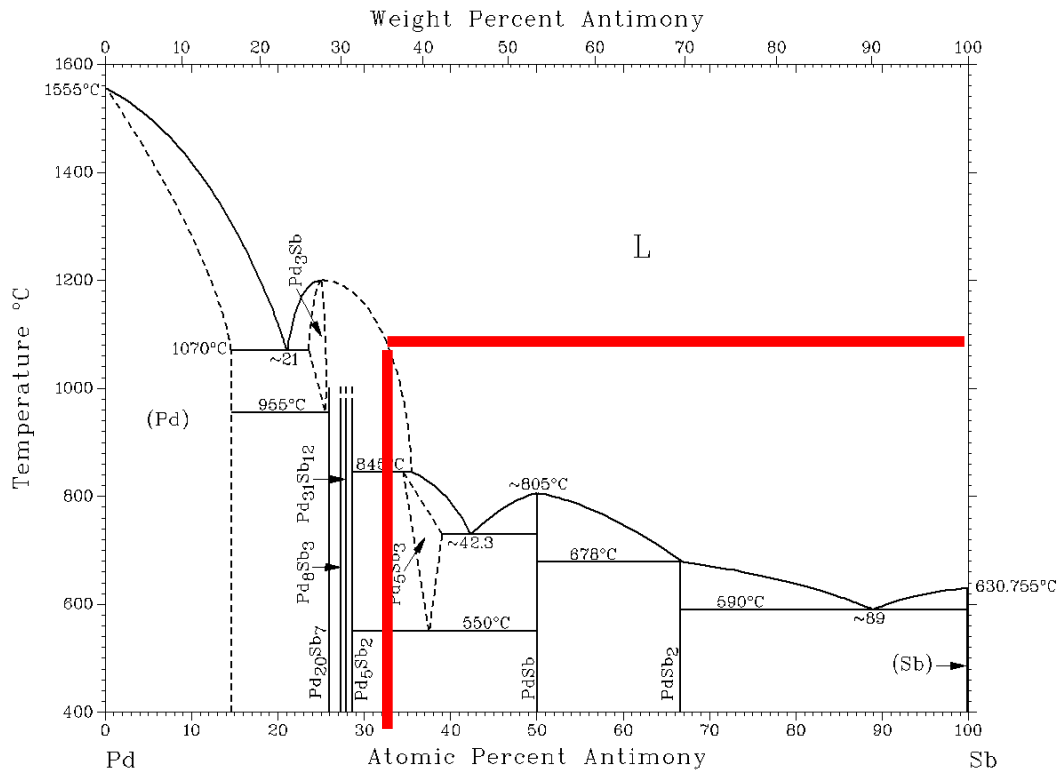
Element	Expected ratio	Final ratio
U	1	1
Ni	1	0,39
Sb	1,33	1,72

~U_3NiSb_5, unreported new phase

$U_3Pd_3Sb_4$

Upper temperature 1100°C

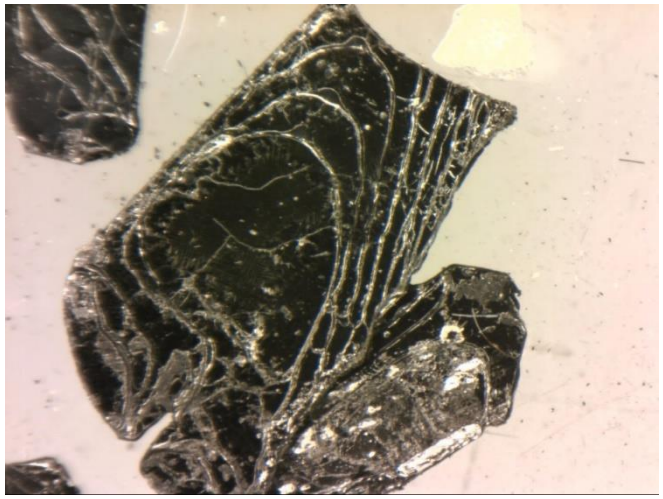


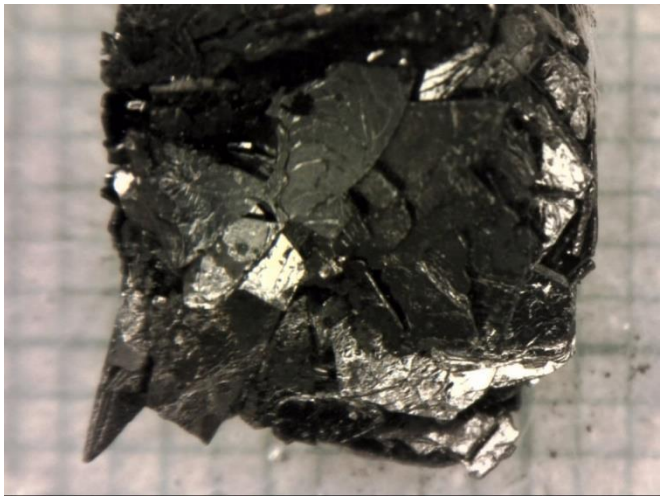


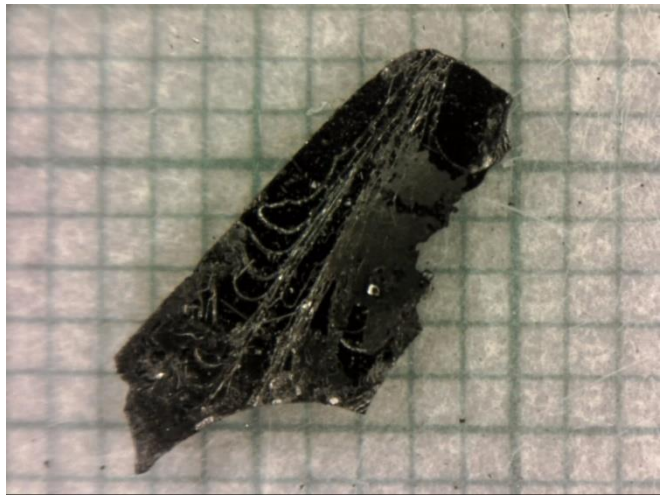
Solubility of Pd in Sb-flux is significantly higher than U

$U_3Pd_3Sb_4$

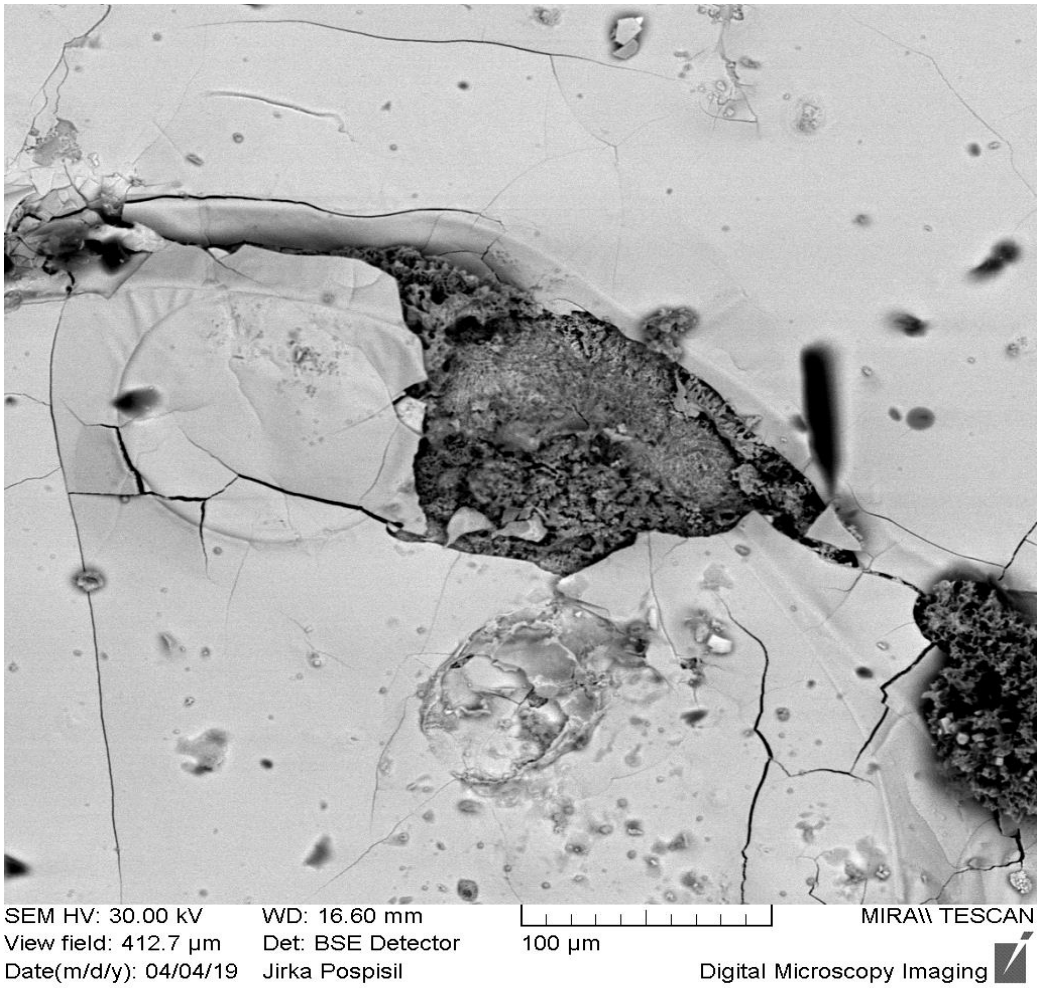
Element	Sb(6N)		Pd(4N5)		U(3N)
Ratio	10	:	2,5	:	1
Mass	4,77 g		1,14 g		0,91 g

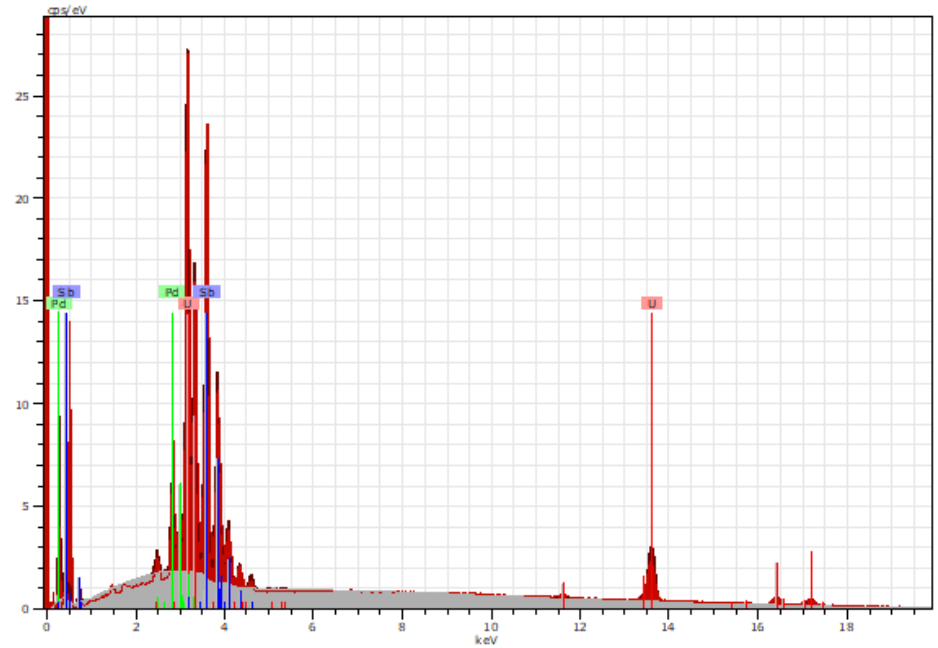






$U_3Pd_3Sb_4$ – Chemical analysis



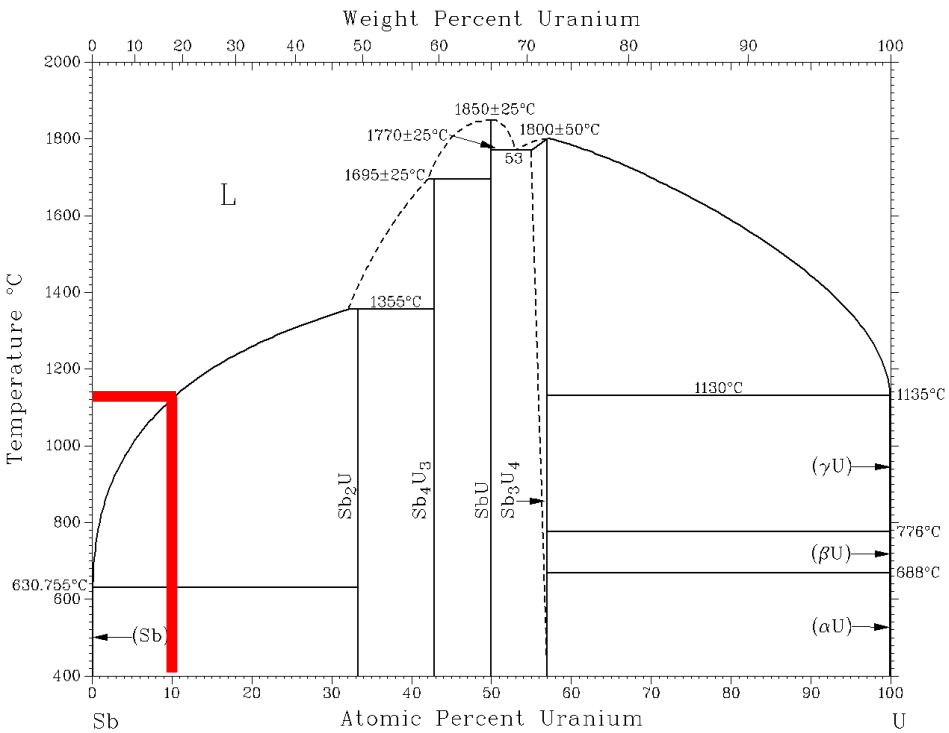


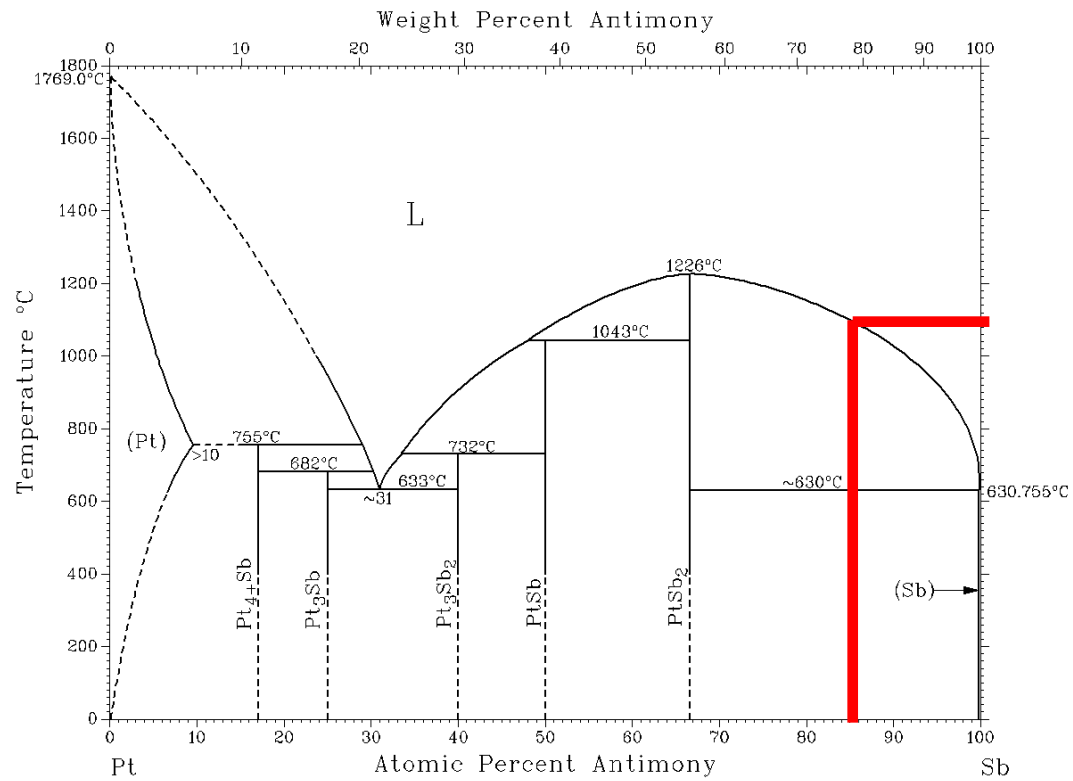
Element	Expected ratio	Final ratio
U	1	1
Pd	1	0,32
Sb	1,33	1,74

~U_3PdSb_5, unreported new phase,
identical composition like Ni case

$U_3Pt_3Sb_4$

Upper temperature 1100°C

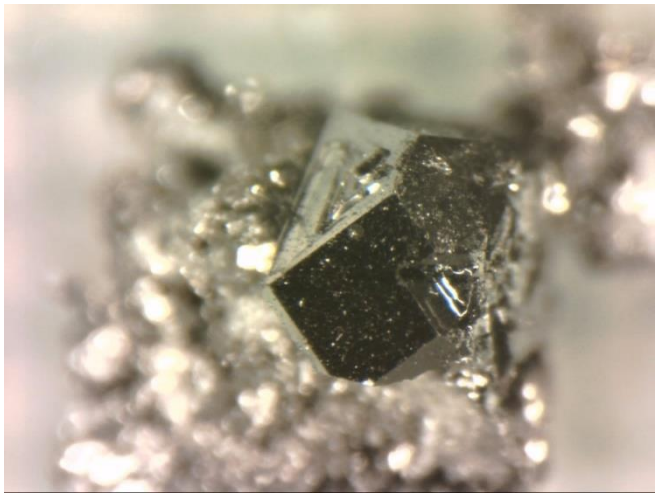


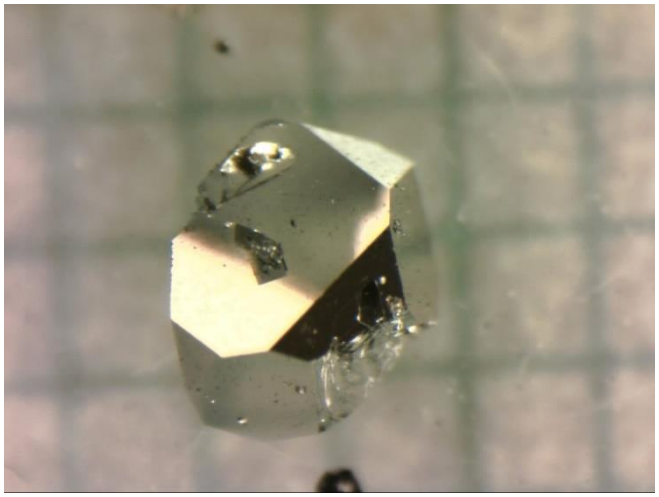


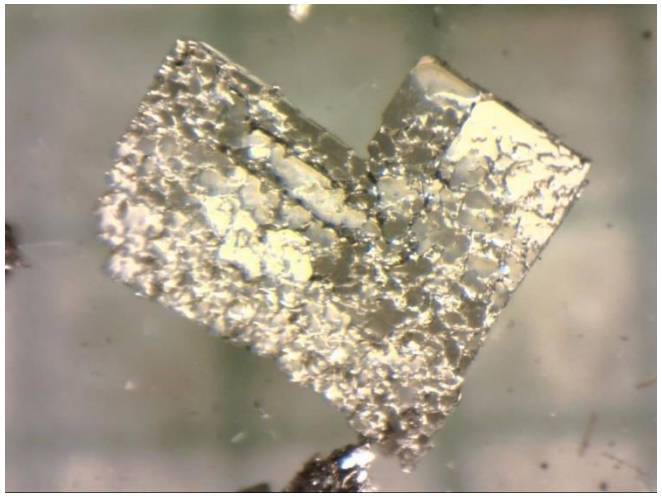
- Solubility of Pt in Sb-flux is very similar to U

$U_3Pt_3Sb_4$

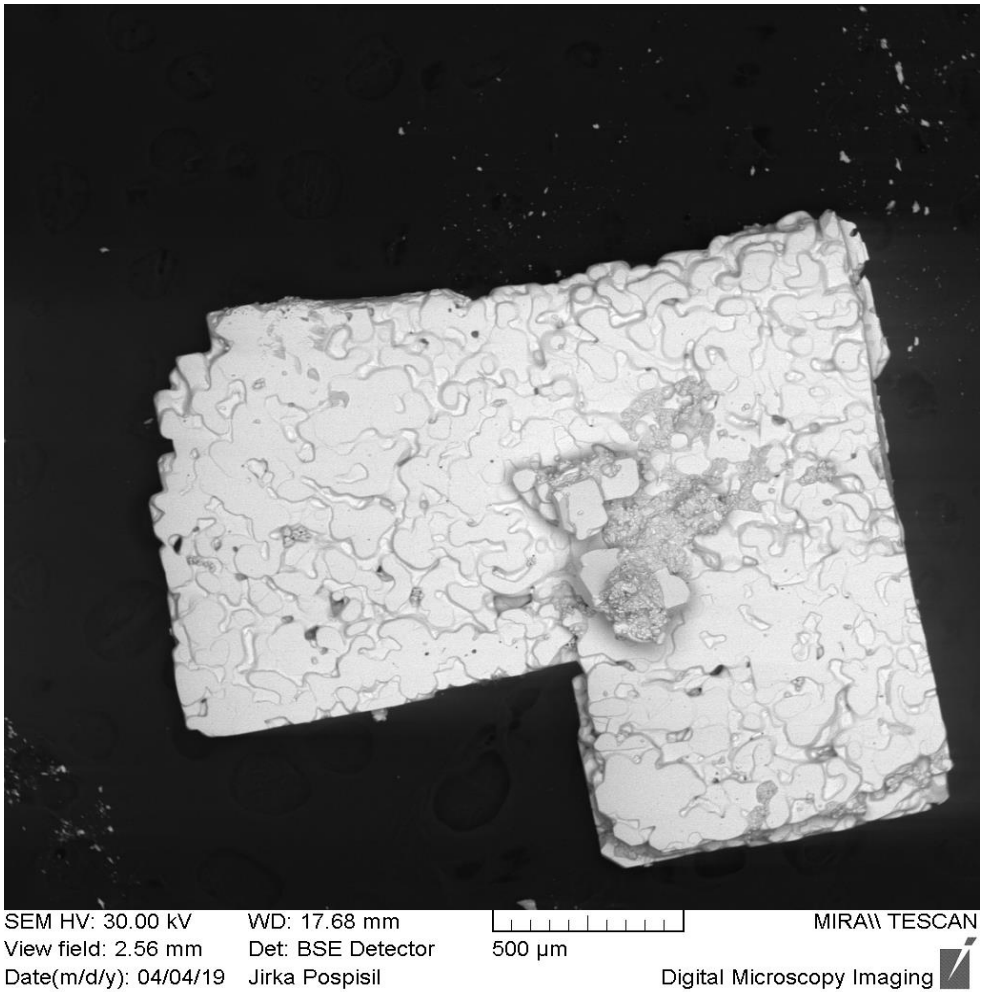
Element	Sb(6N)		Pt(4N5)		U(3N)
Ratio	10	:	1	:	1
Mass	4,13 g		0,66 g		0,79 g

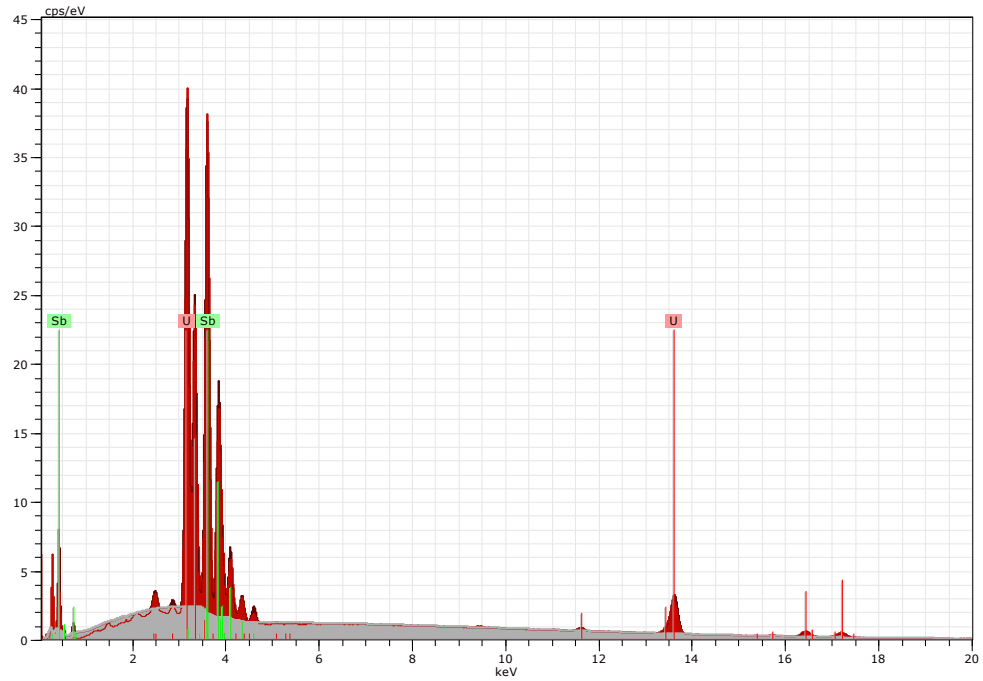






$U_3Pt_3Sb_4$ - Chemical analysis

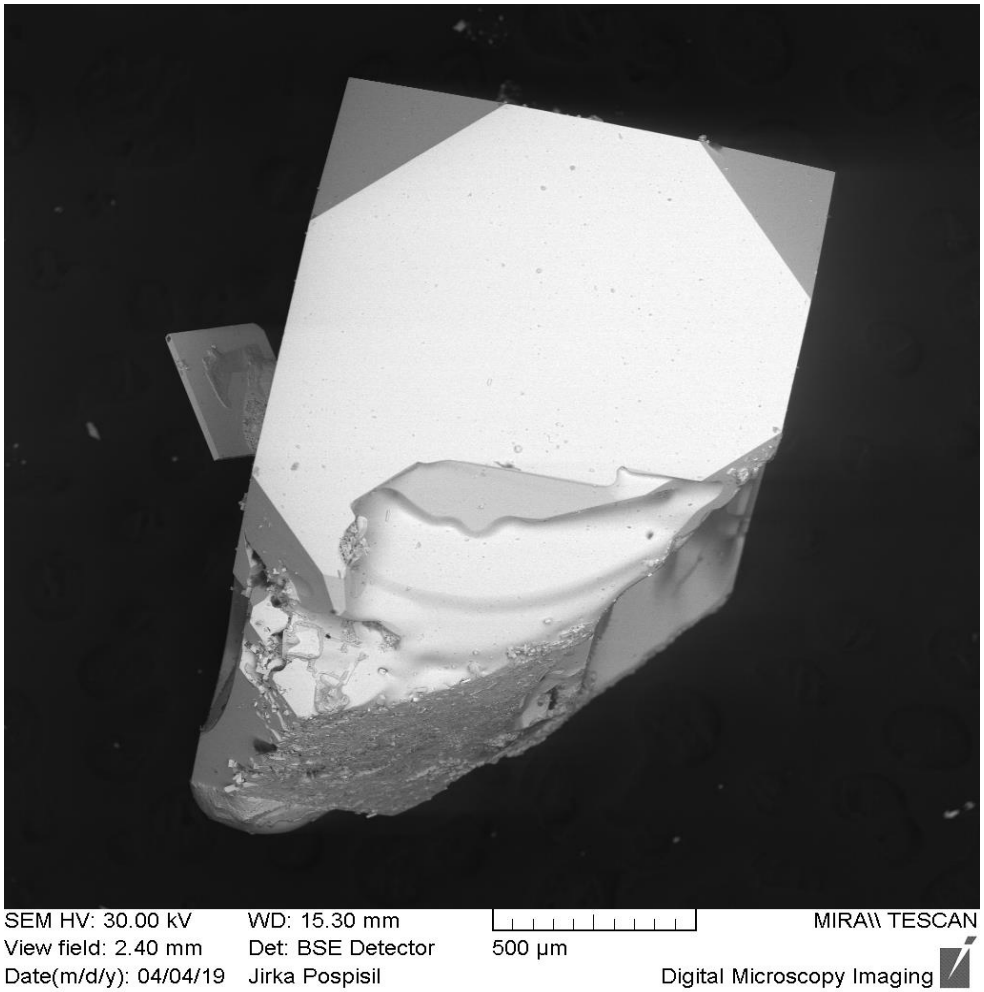


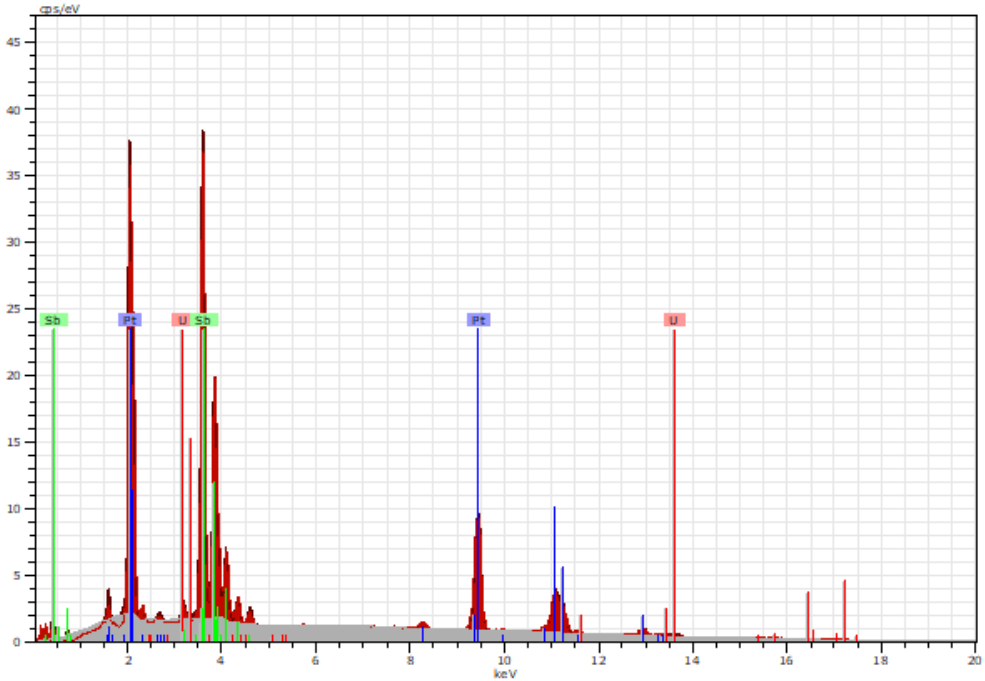


Element	Expected ratio	Final ratio
U	1	1
Pt	1	0
Sb	1,33	1,83

~USb_2, binary well known phase
Antiferromagnet T_N = 203 K

$U_3Pt_3Sb_4$ - Chemical analysis

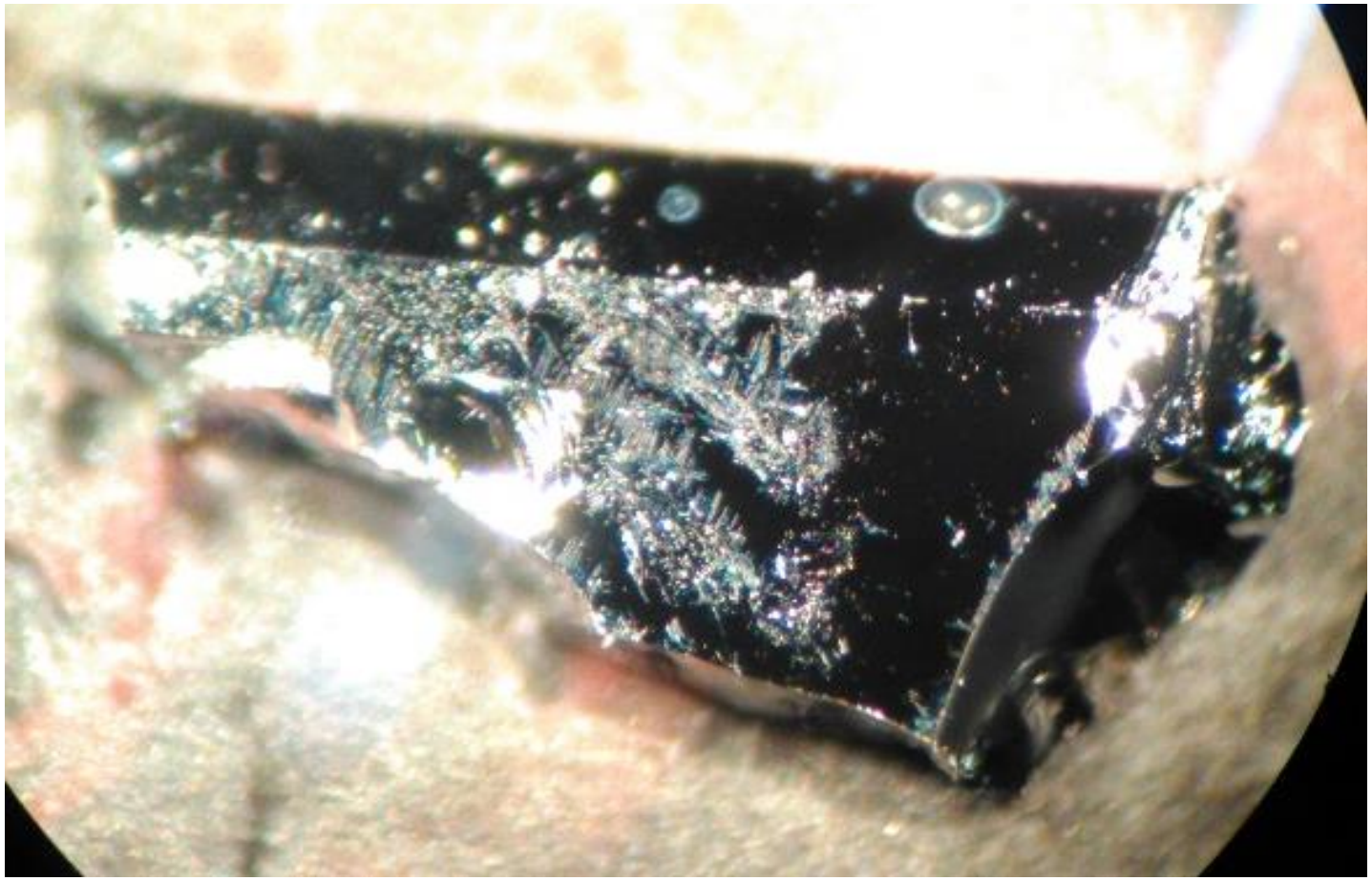




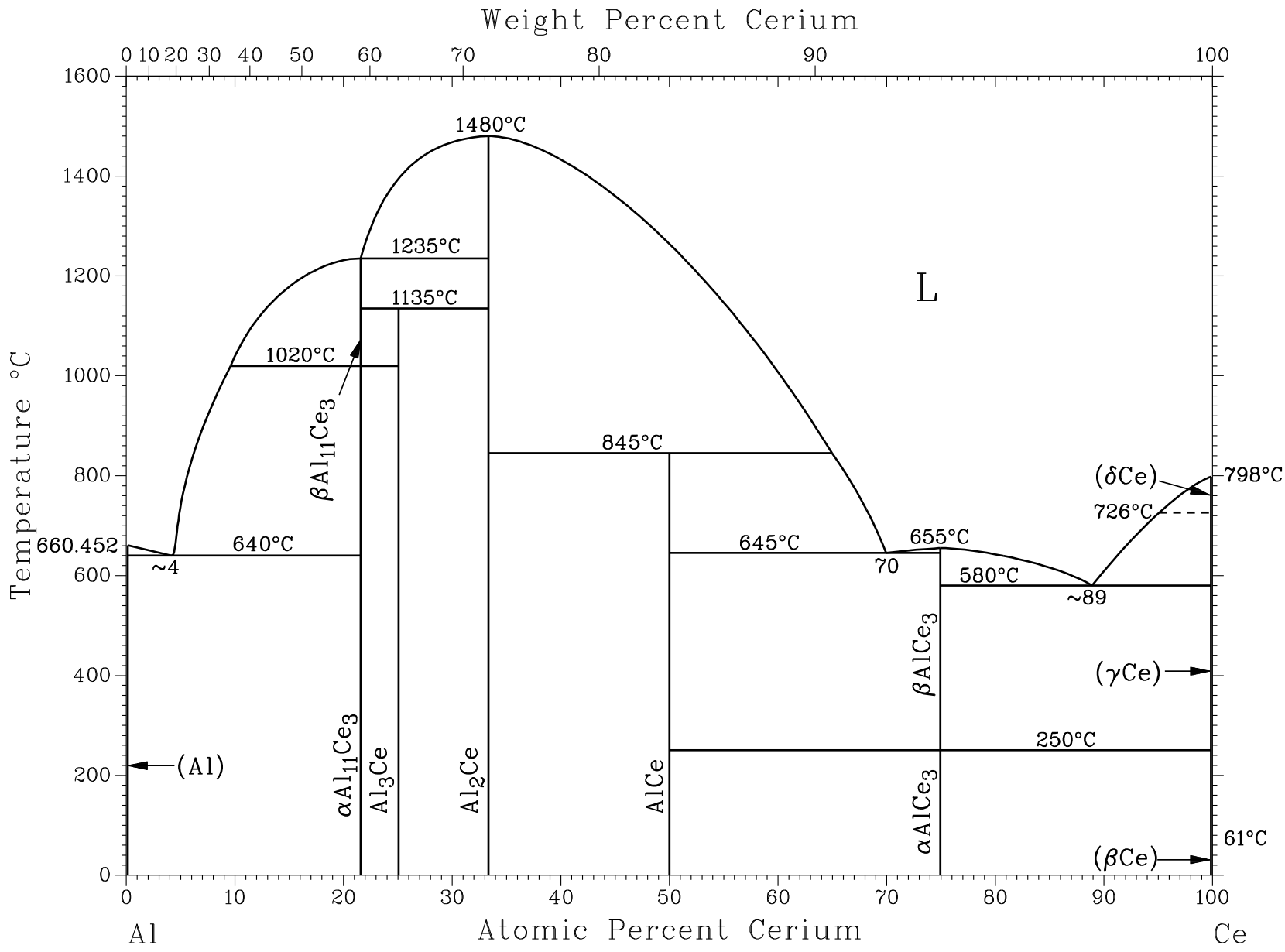
Element	Expected ratio	Final ratio
U	1	0
Pt	1	1
Sb	1,33	2,03

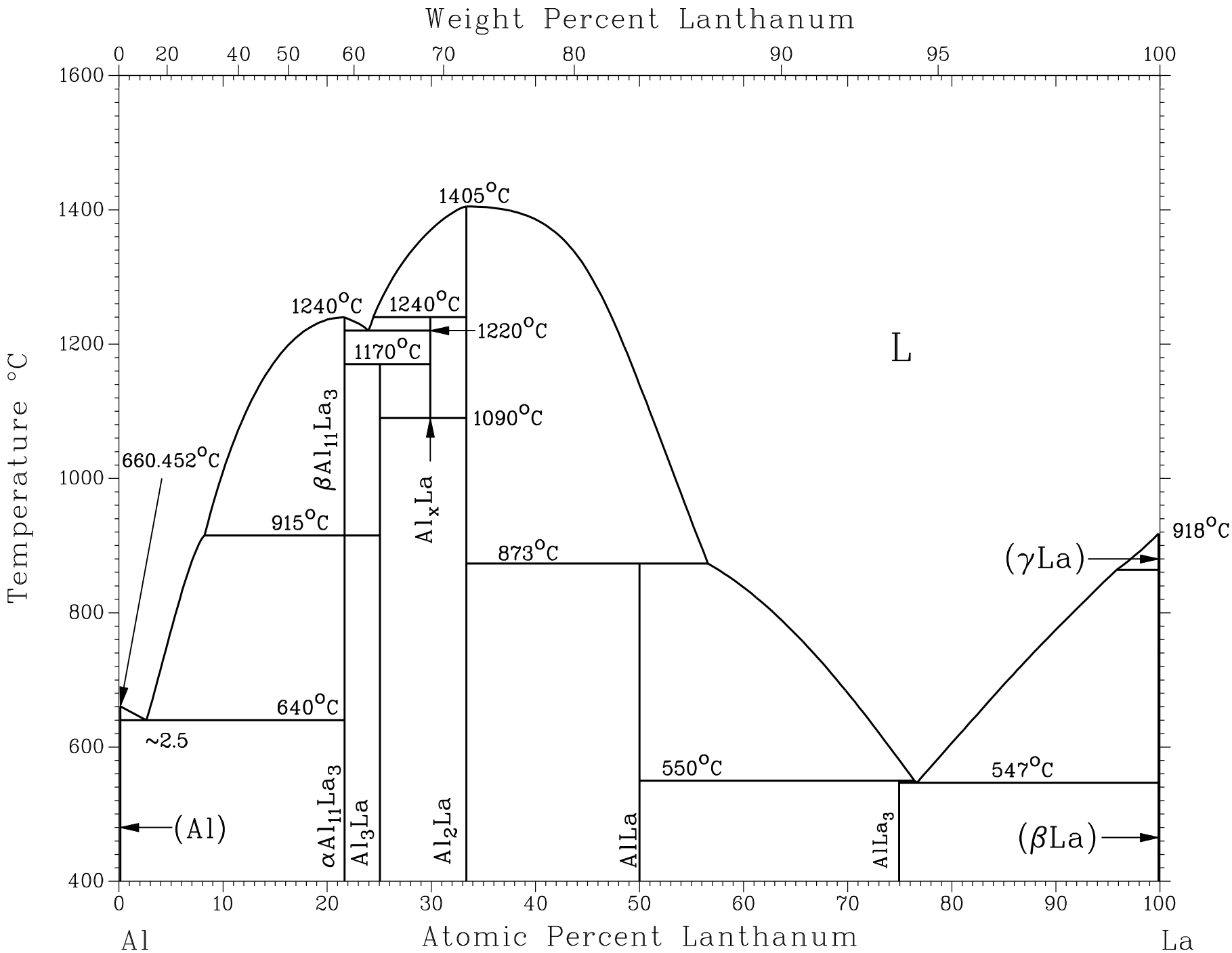
~$PtSb_2$, binary well known phase Narrow-band semiconductor (investigated in the 1960's).

Ce_3Al_{11}



Assessed Al-Ce phase diagram.




Weight Percent Lanthanum
0
20
40
50
60
70
80
90
95
100
1600
1400
1200
1000
800
600
400
Temperature °C
1405°C
1240°C
1240°C
1220°C
1170°C
1090°C
L
660.452°C
915°C
873°C
918°C
(γLa)
640°C
~2.5
550°C
547°C
(Al)
(βLa)
βAl11La3
αAl11La3
Al3La
AlxLa
Al2La
AlLa
AlLa3
0
10
20
30
40
50
60
70
80
90
100
Al
Atomic Percent Lanthanum
La